

Recap 30/9-2025: Two competing events: (Q and decay x/τ)

$$\frac{dx}{dt} = Q - \frac{x}{\tau}$$

- Select time step for increase in x , $\Delta t_Q = -\ln(\text{ran}_1)/Q$ by choosing a random number $\text{ran}_1 \in [0, 1]$.
- Select time step for decrease in x , $\Delta t_x = -\ln(\text{ran}_2)\tau/x$ by choosing a random number $\text{ran}_2 \in [0, 1]$.
- If $\Delta t_Q < \Delta t_x$ then update $x = x + 1$ and set time counter $t = t + \Delta t_Q$.
- If $\Delta t_Q > \Delta t_x$ then update $x = x - 1$ and set time counter $t = t + \Delta t_x$.

Recap 30/9 2025, SIR models:

$$\begin{aligned}\frac{dS}{dt} &= -\lambda \cdot S \cdot I \\ \frac{dI}{dt} &= \lambda \cdot S \cdot I - \gamma I \\ \frac{dR}{dt} &= \gamma \cdot I\end{aligned} \quad \Rightarrow \quad R_0 = \frac{\lambda}{\gamma},$$

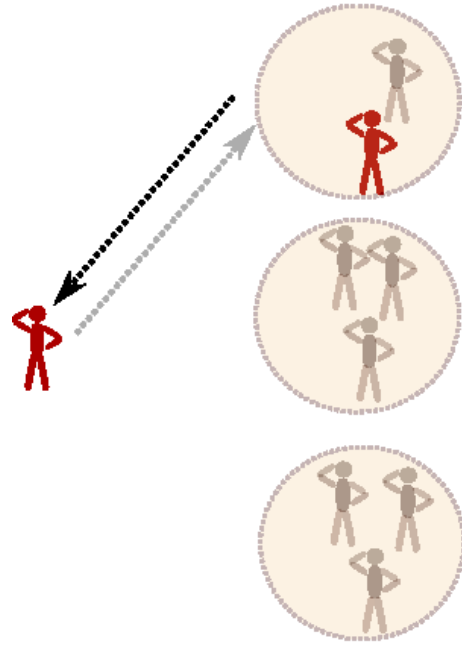
And infecting R_0 people takes $1/\gamma$ time

Growth rate = $dI/dt = (\lambda - \gamma) I$ (For $S \sim 1$)

$$\frac{d \ln(S)}{dR} = -\lambda/\gamma = -R_0 \Rightarrow S(\infty) = e^{-R_0(1-S(\infty))}$$

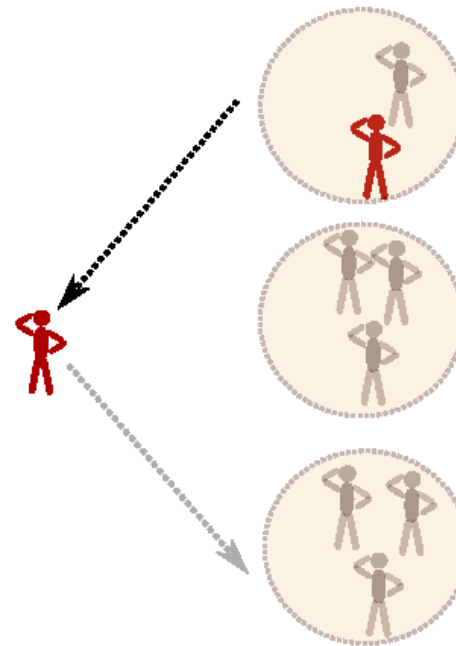
Recap: Agent based contra SIR=random mixing

Reality: Each man return to same home every day



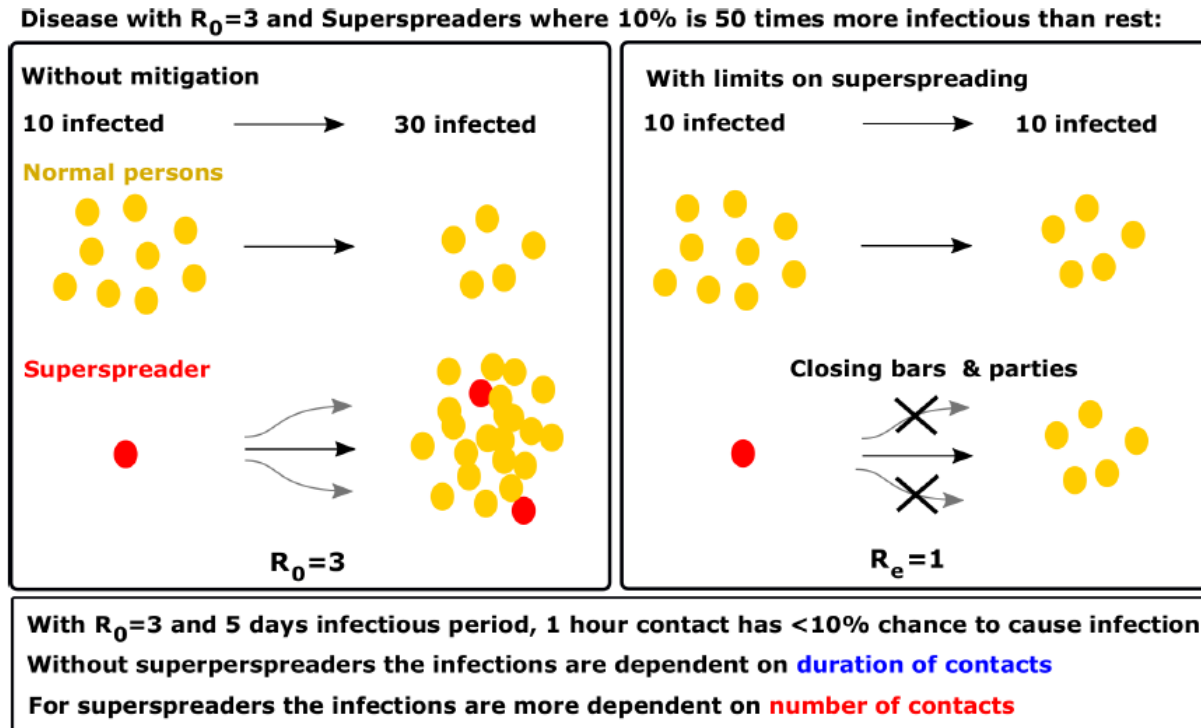
Here can only infect up to 2 new susceptibles

SIR simplification: Each man return to a new home every day



Here he has unlimited infection potential (one for each timestep)

- An example where Agent-based models become crucial.



Have to deal with limits from **social Networks**, and **Finite time**

.....Important when discussing mitigation=social confinement

Agent-based model of SIR model:

(Quenched contacts for individuals, ER network)

Each timestep, $[t, t+dt]$

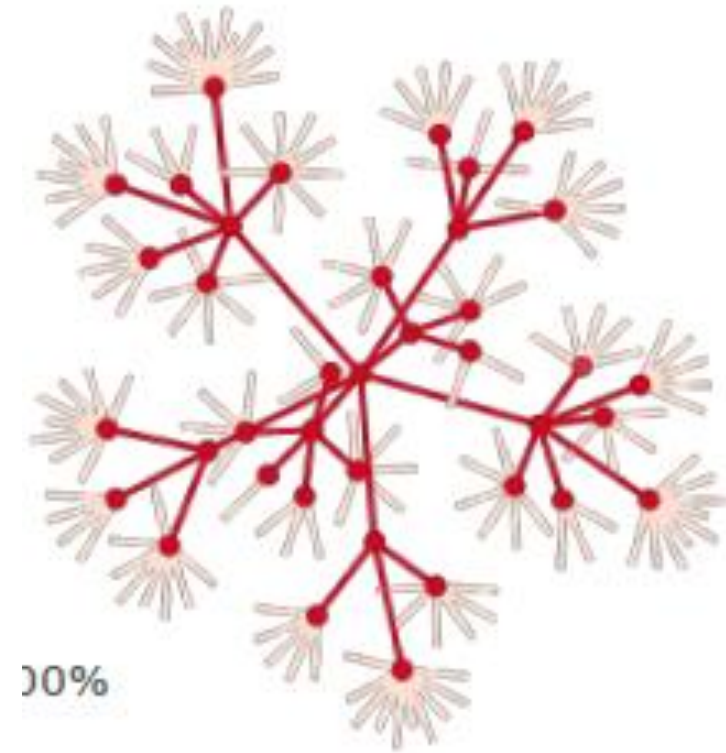
Select all infectious agents,

One by one, call one j .

Let j become R with probability γdt ,

With probability λdt select
a random neighbor k in j 's social network.
If this agent k is S it becomes infectious
($S \rightarrow I$). Otherwise, nothing happens.

$$\begin{aligned}\frac{dS}{dt} &= -\lambda \cdot S \cdot I \\ \frac{dI}{dt} &= \lambda \cdot S \cdot I - \gamma I \\ \frac{dR}{dt} &= \gamma \cdot I\end{aligned}$$



Notice that social time of everybody is here $2 dt$ (in principle, you are meeting people twice)
Notice that you cannot be infected if none of your neighbors are infectious.

Networks

Kim Sneppen

September 30, 2025

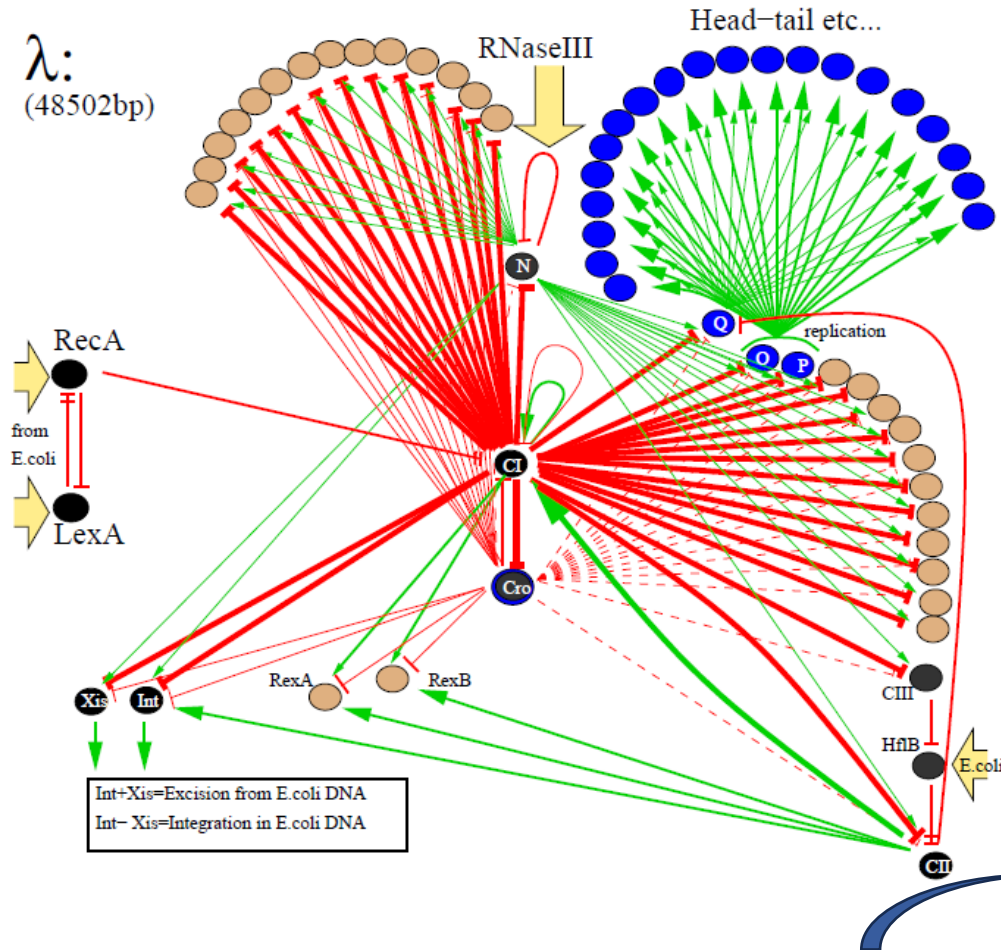
- **Heterogeneity:** Systems with distinctly different components
- **History:** A real network does not appear by random assignment of links, but self-organizes over a long time period. Networks are thus useful for **systems with a separation of time scales**, with a dynamics on the network that occurs repeatedly and on a fast timescale. Network rewiring is in contrast much slower.
- **Distribution & containment** of information, energy or material.

Not between
Atoms.
Rather between
complex entities

SOC
like

6 degrees of separation....
Rather, it is more about limiting access, as you will
Feel when you try to access nodes higher in social hierarchy.

Links are directed, and often with a sign



Networks have meaning beyond statistics.

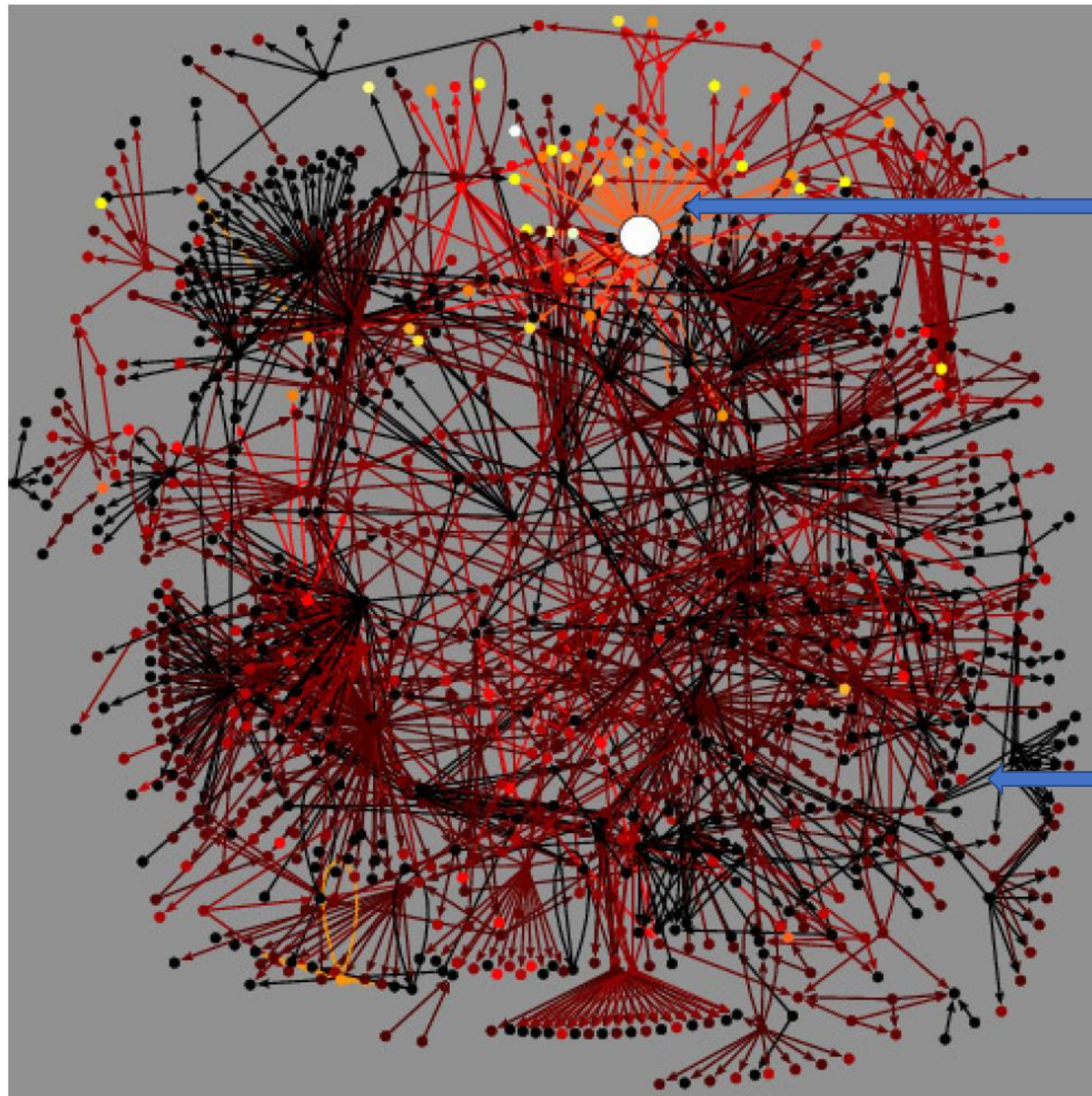
(Which is what we do in this course)

Notice that Some nodes are much more connected than others...

This was a biological network:

- 1) Between very different proteins,*
- 2) evolved over millions of years with a dynamic timescale of hours (inside each cell)*
- 3) And they were about information transfer (regulatory networks inside a cell)*

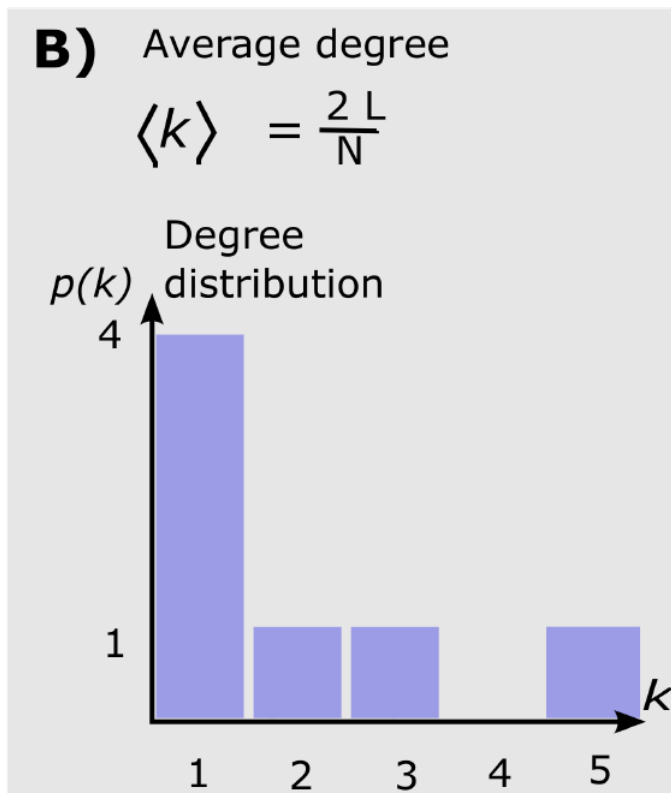
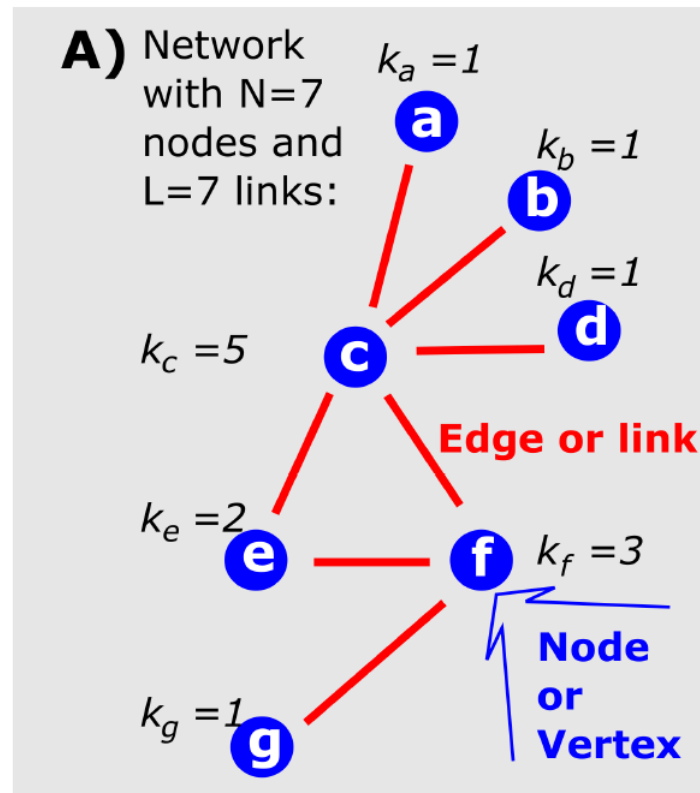
In addition, often a network is a list of **potential** interactions that are active sometimes, but not always



Some parts active

Other parts passive
...
Represent potential
not always present.

Basic Concepts: The node degree=connectivity → Distribution $p(k)$



Basic Concepts, Erdos-Reynei Network:

- N nodes, L Links, Randomly distributed →

$$\langle k \rangle = \frac{2L}{N} = p(N-1)$$

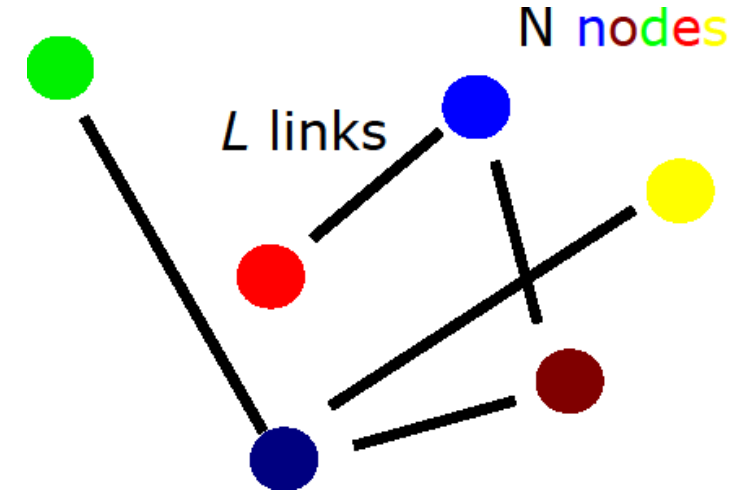
Distributed randomly around mean $p=2L/(N(N-1))$:

$$P(k) = \frac{(N-1)!}{k!(N-1-k)!} \cdot p^k \cdot (1-p)^{(N-1-k)}$$

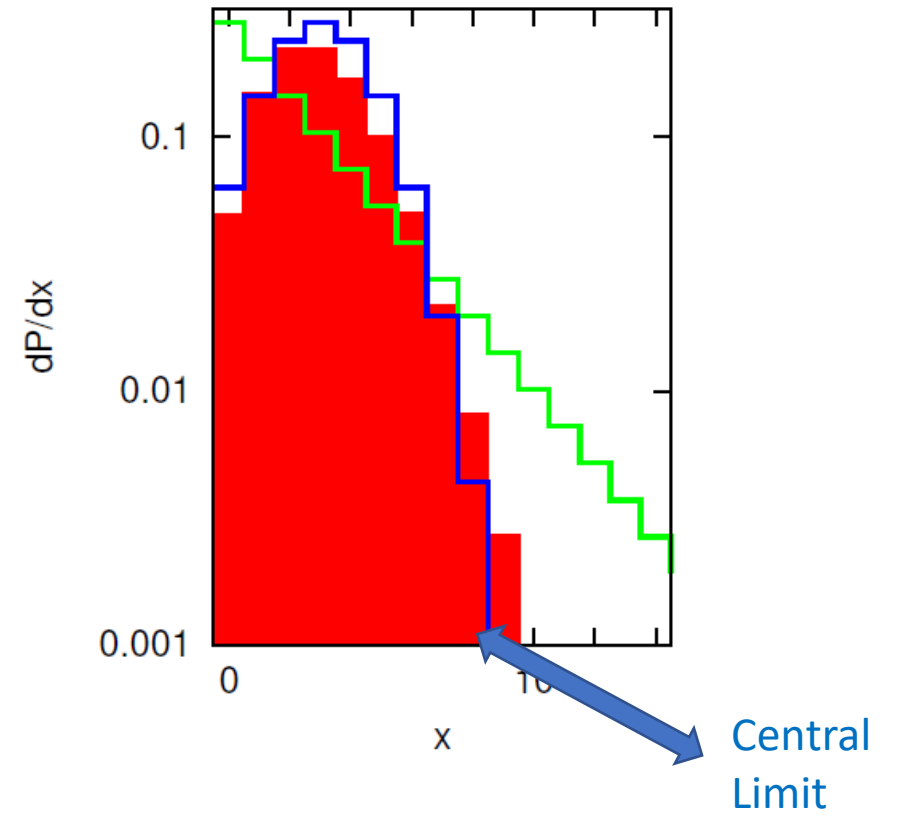
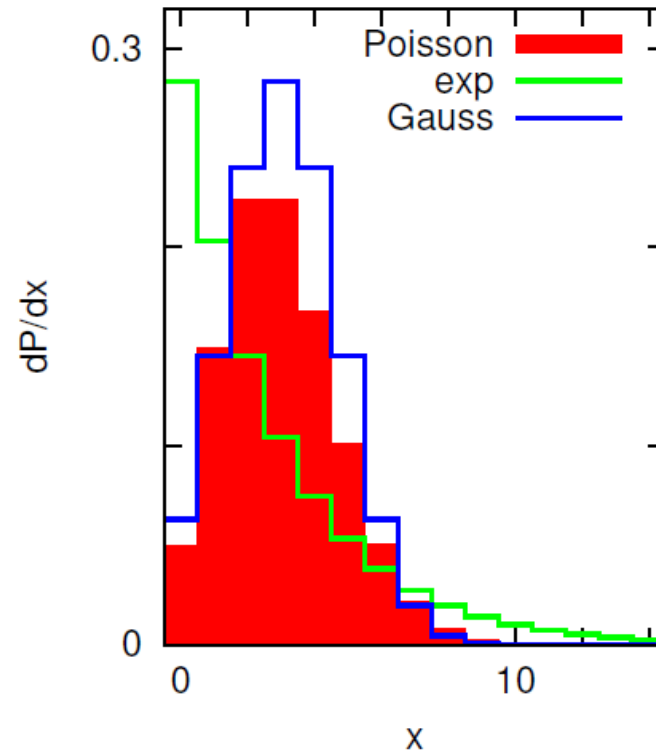
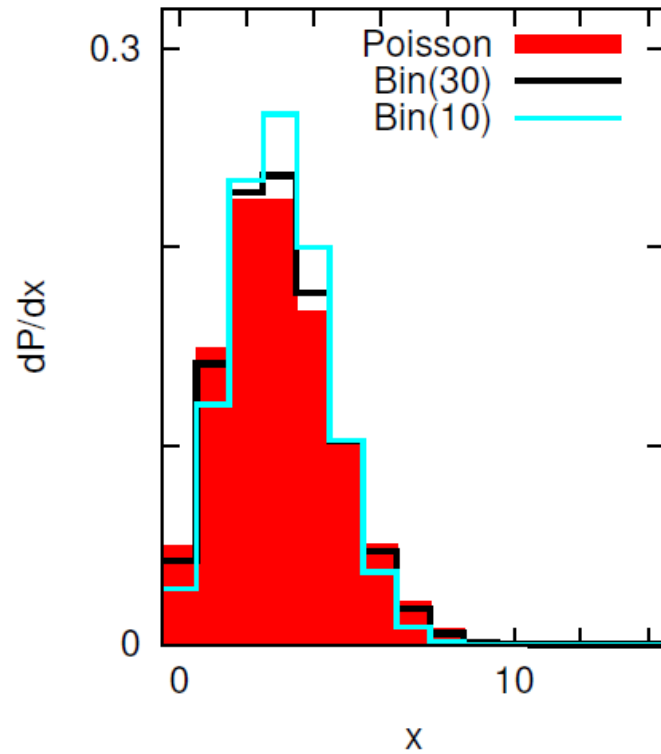
Large N limit → Poisson distributed:

$$P(k) = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}$$

$$var = \langle k^2 \rangle - \langle k \rangle^2 = \langle k \rangle, \text{ and thus } \langle k^2 \rangle = \langle k \rangle \cdot (\langle k \rangle + 1)$$



Reminder/comparison:



Basic concepts: Distance in & Diameter of Network

- $d(i,j)$ minimal number of links needed to connect node i and node j

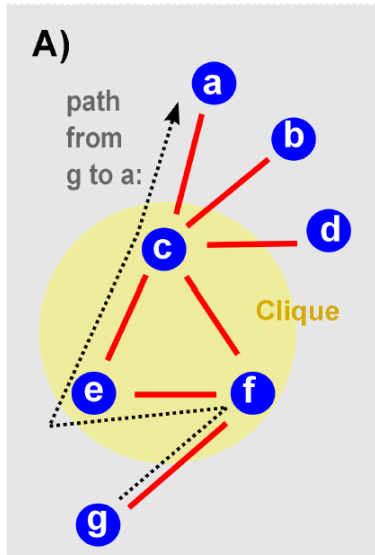
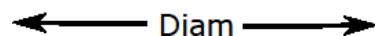
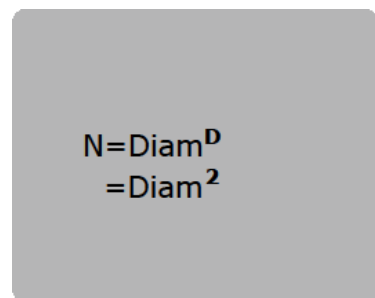


Figure show one path, but the shortest go directly from f to c.

$$\text{number}(\text{nodes within } d) = k \cdot \sum_{d'=1}^d (k-1)^{(d'-1)} \sim k \cdot (k-1)^{d-1}$$

(only last layer count, at least when $k > 2$)

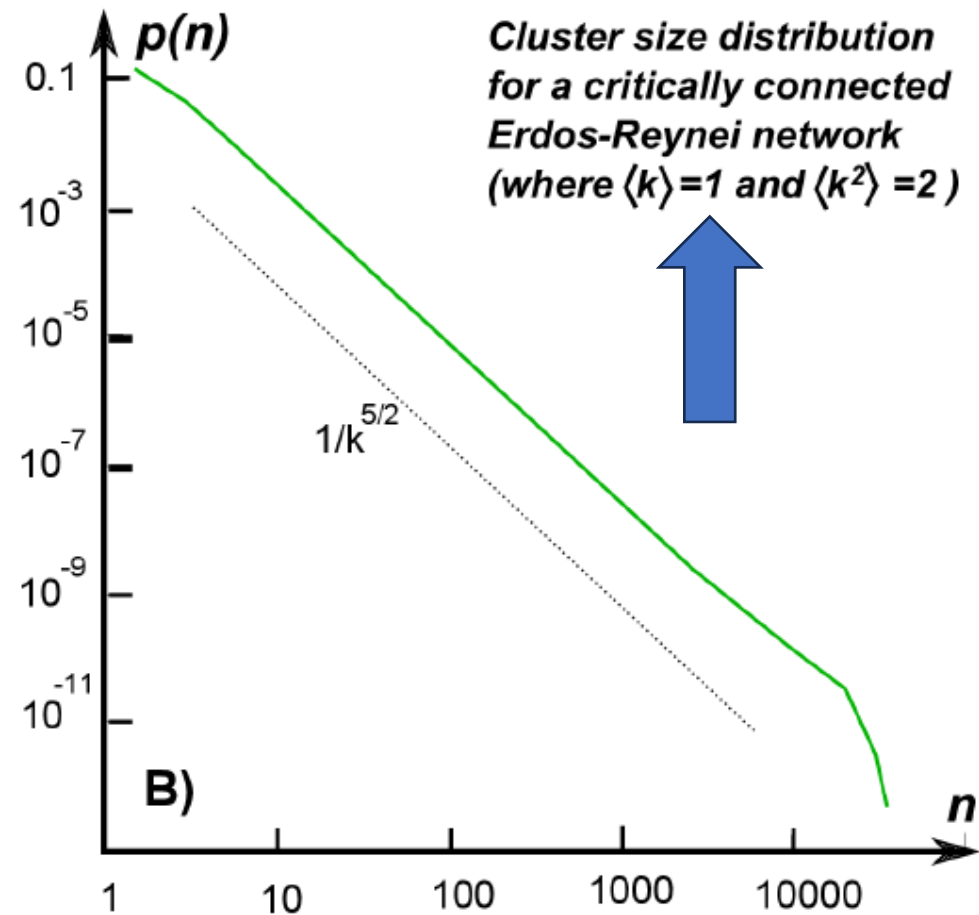


Diam=maximal of all distances→

All nodes $N < k^{\text{Diam}} \rightarrow \text{Diam} > \sim \ln(N)/\ln(k)$

i.e. a **random network have infinite dimension (like Bethe lattice)**
(any finite dimension D would have $N < \text{Diam}^D$)

Cluster sizes at criticality in Erdos-Reyni network = as critical Bethe lattice



Amplification factor: $A=1$
... cluster size same as
in Bethe lattice with $p=1/2$.

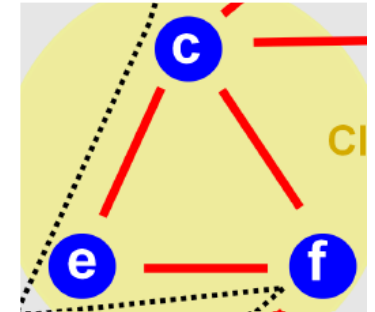
Cliquishness of a node:

$$C_i = \frac{\text{number of pairs of neighbors to } i \text{ connected to each other}}{k_i \cdot (k_i - 1)/2}$$

C=average Cliquishness:

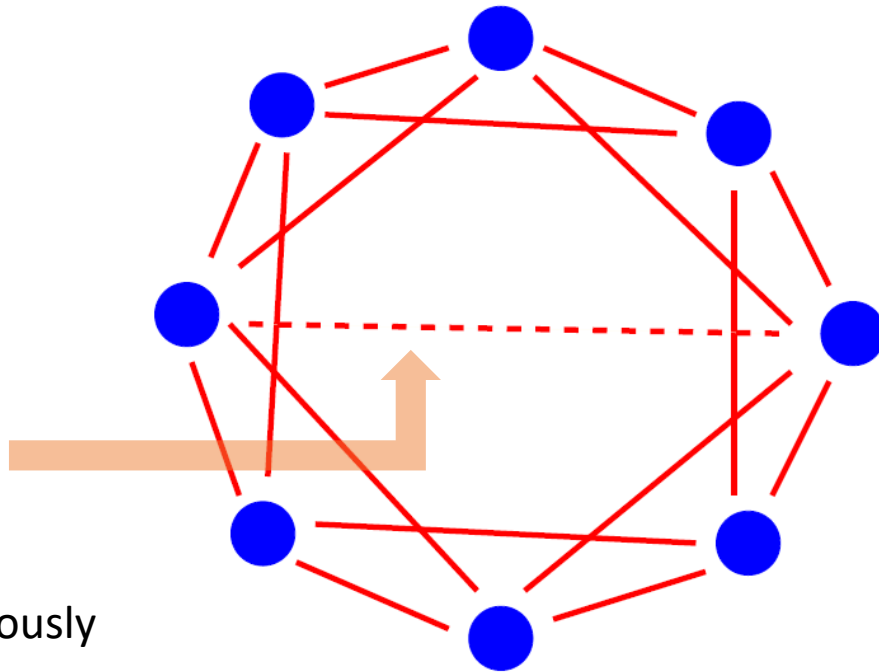
For an Erdos-Reynei network: $C = p \sim \frac{\langle k \rangle}{N}$

Mini tutorial: How many triangles (loop 3 cycles) are there in an Erdos-Reynei network as function of network size N



Small world network: Many cliques, and short distances ($\sim \log(N)$)

SMALL WORLD = many cliques and small diameter (order $\log(N)$)



A few non
Local links
shorten all
distances enormously

Network with rather large
cliquishness

$C \gg C(\text{random expected})$

here $C = 3/6$ (except ----
and with few additional
links (---), it also have

rather small diameter.

Therefore SMALL WORLD

Betweenness: A way to quantify links/nodes that are most central/essential in networks

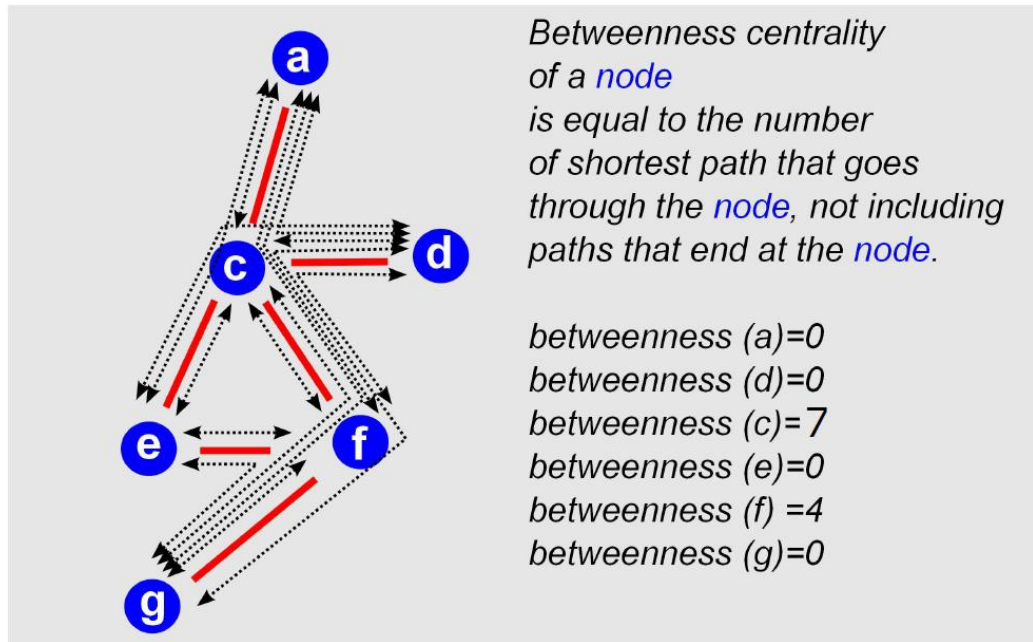


Figure 2.6: Illustrating the betweenness of a node. *Betweenness* measures the communication traffic that runs through the node, assuming that all pairs of nodes send messages to each other and always use the shortest path between them [27, 28].

Tutorial:
Draw a network where one
node has high betweenness,
But low connectivity

Networks & Sparse matrices

Bi-directional:

In network form:



In matrix form:

$$\mathbf{A} \mathbf{v}_1 = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$\begin{matrix} 1 & 2 & 3 \\ \text{to} & \text{to} & \text{to} \\ j & j & j \end{matrix}$

Directed:

In network form:



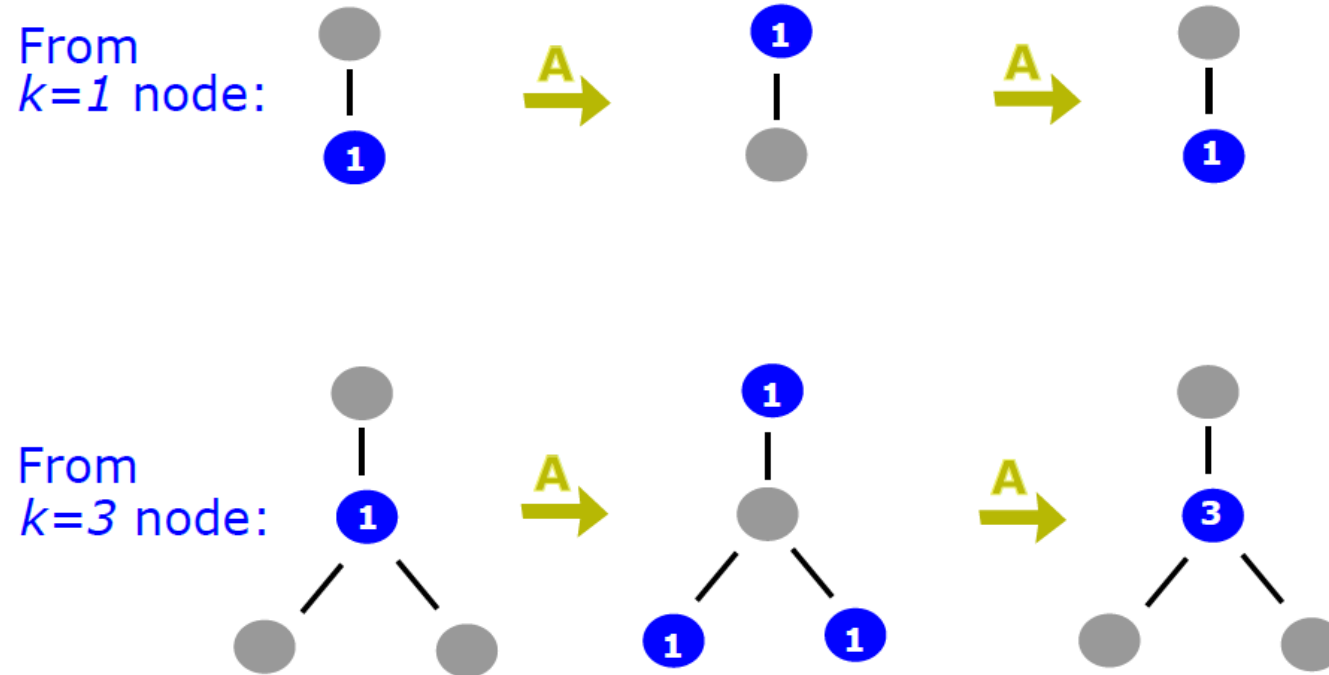
In matrix form:

$$\mathbf{B} \mathbf{v}_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$\begin{matrix} 1 & 2 & 3 \\ \text{to} & \text{to} & \text{to} \\ j & j & j \end{matrix}$

Matrix multiplication and connectivity

Two step spreading (A^2)



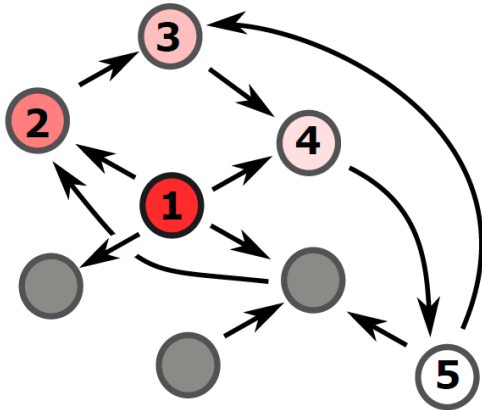
One start with 1 unit at central nodes: after 2 iterations the node has k units.

Number of 3 loops in a network

$$n(\triangle) = \frac{1}{6} \cdot \text{trace} (\mathbf{A}^3)$$

Provided that A does not have self loop (diagonal of **A=0**)

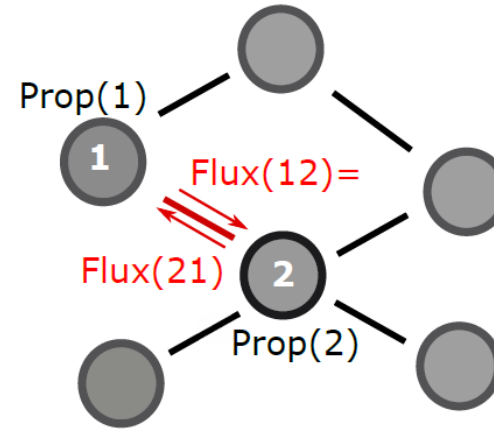
Steady-state random walks in networks;



A random walk on a network visits each node proportional to its in-degree.

Here directed network as explored by Google search algorithm.

Numbers represent subsequent iterations
As the walker select a random exit at each node.



Detailed balance:
 $\text{Flux}(12) = \text{Flux}(21)$
 $\text{Prop}(1) \times 1/k_1 = \text{Prop}(2) \times 1/k_2$
 $\text{Prop}(1)/2 = \text{Prop}(2)/4$
 Thus it is double as likely to visit node 2 as visit node 1

$$\frac{v_i^1}{k_i} = \frac{v_j^1}{k_j} = \text{constant}$$

Random walk matrix:

Transfer matrix:

$$\mathbf{T} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & \\ T_{31} & \end{bmatrix} = \begin{bmatrix} 0 & k_2^{-1} & k_3^{-1} \\ k_1^{-1} & 0 & \\ k_1^{-1} & & 0 \end{bmatrix}$$

for the case where $A_{12}=A_{21}=1$, $A_{13}=A_{31}=1$

Because a walker select on of k neighbors

only non-zero elements when $A_{ij}=1$

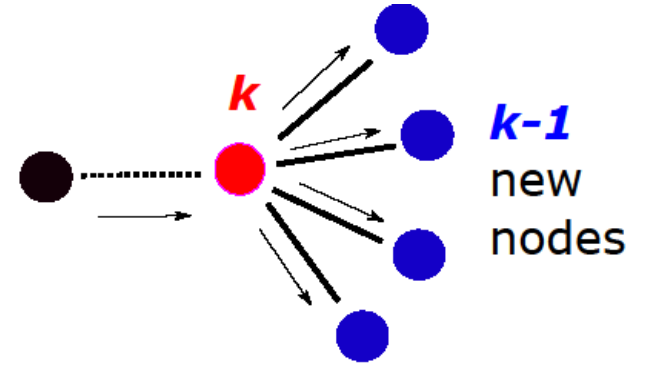
$$\mathbf{T} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & k_2^{-1} & k_3^{-1} \\ k_1^{-1} & 0 & \\ k_1^{-1} & & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ k_1^{-1} \\ k_1^{-1} \end{bmatrix}$$

corresponding to spreading $1/k$ to each neighbor

**Non symmetric, but anyway have real eigenvalues!!!
(not proven here)**

K. A. Eriksen, I. Simonsen, S. Maslov, and K. Sneppen. Modularity and extreme edges of the interet. *Phys. Rev. Lett*, 90:148701, 2003.

Amplification factor:



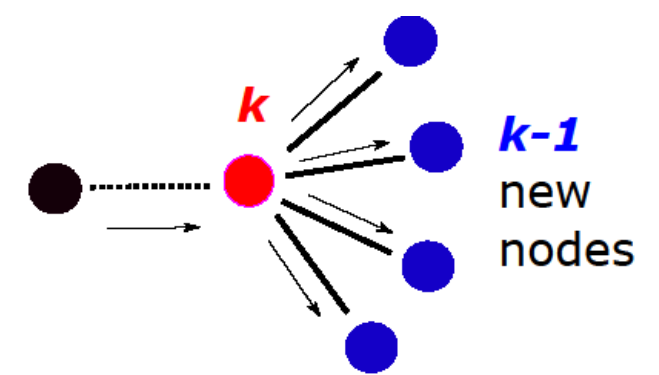
$$\mathcal{A} = \frac{\int k(k-1)n(k)dk}{\int kn(k)dk} = \frac{\langle k(k-1) \rangle}{\langle k \rangle} = \frac{\langle k^2 \rangle}{\langle k \rangle} - 1$$

Propability to select **red node** proportional to **k**
Amplification factor **k-1**

→ Your friends have more you than you,
The Amplification factor is often larger than average k
Poisson distribution with $\langle k \rangle = 1 \rightarrow \langle k^2 \rangle = 2 \rightarrow \mathcal{A} = 1$

$\mathcal{A} = 1$ correspond to critical branching.

Amplification factor:



Probability to select **red node** proportional to k
Amplification factor $k-1$

$$\mathcal{A} = \frac{\int k(k-1)n(k)dk}{\int kn(k)dk} = \frac{\langle k(k-1) \rangle}{\langle k \rangle} = \frac{\langle k^2 \rangle}{\langle k \rangle} - 1$$

→ Your friends have more friends you than you:

$$\langle k^2 \rangle / \langle k \rangle > \langle k \rangle$$

The Amplification factor is often larger than average k

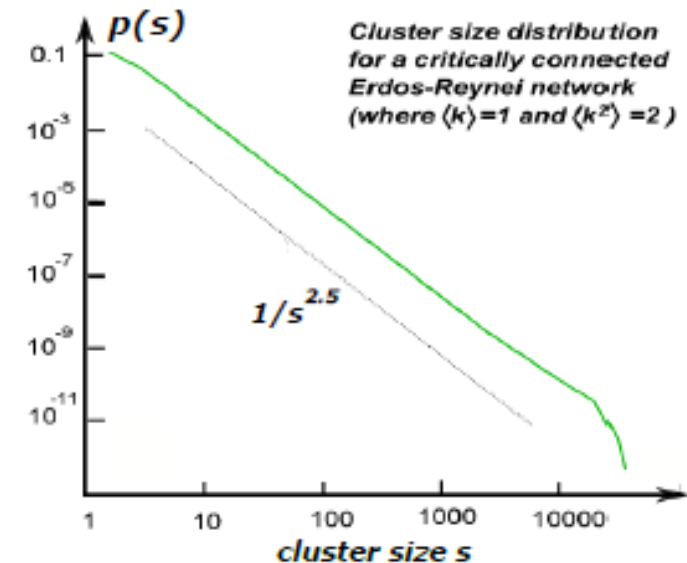
Poisson distribution with $\langle k \rangle = 1 \rightarrow \langle k^2 \rangle = 2 \rightarrow A = 1$

$A = 1$ correspond to critical branching.

Erdos-Reynei with $\langle k \rangle = 1$ is critical....

(like a critical percolation cluster in Bethe Lattice)

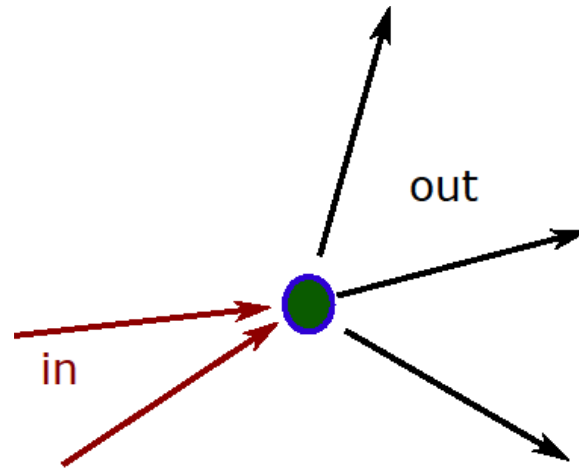
Probability of cluster sizes



$$var = \langle k^2 \rangle - \langle k \rangle^2 = \langle k \rangle, \text{ and thus } \langle k^2 \rangle = \langle k \rangle \cdot (\langle k \rangle + 1)$$

Mini tutorial: Write the above expression for a directed network, using nodes in degree (k_{in}) and out degree (k_{out}). Consider situation where in and out degrees are uncorrelated.

Mini tutorial: Write the above expression for a directed network, using nodes in degree (k_{in}) and out degree (k_{out}). Consider situation where in and out degrees are uncorrelated.



$$A = \frac{\int k_{out} \cdot p(k_{out} | k_{in}) dk_{out} \cdot k_{in} n(k_{in}) dk_{in}}{\int k_{in} n(k_{in}) dk_{in}}$$

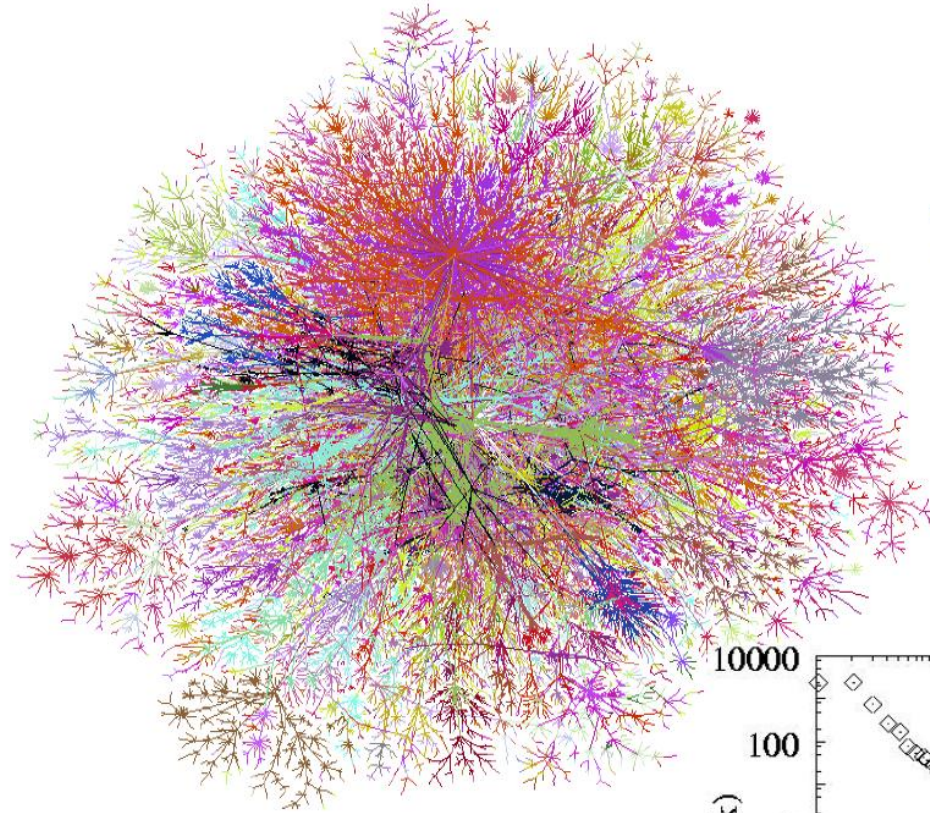
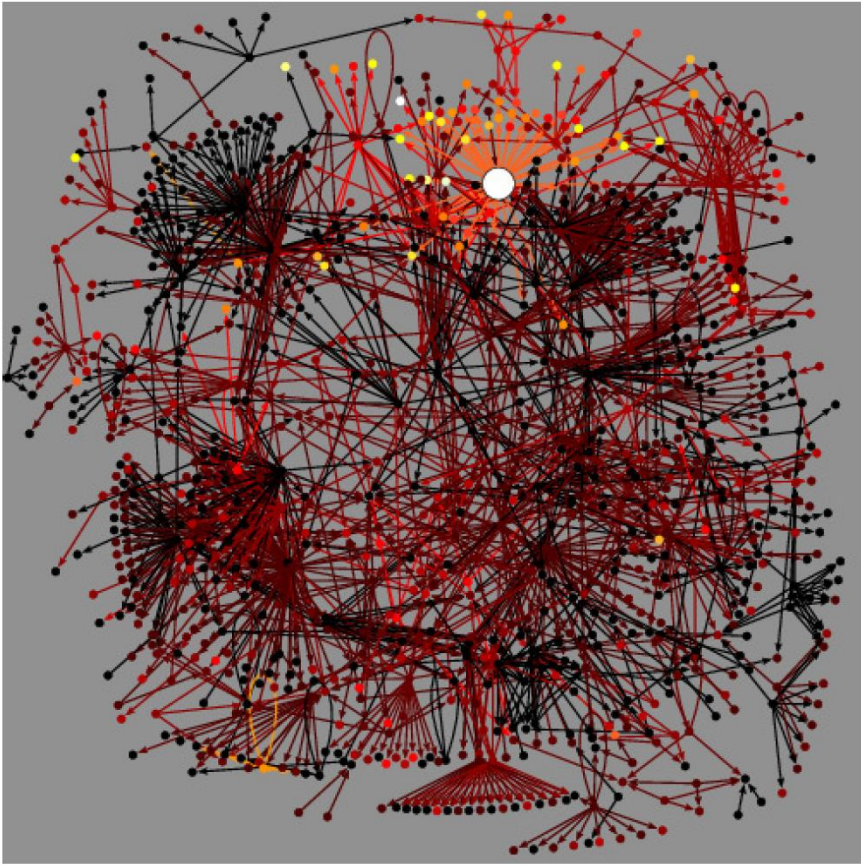
Scale-free networks:

$$n(k) \propto \frac{1}{k^\gamma}$$

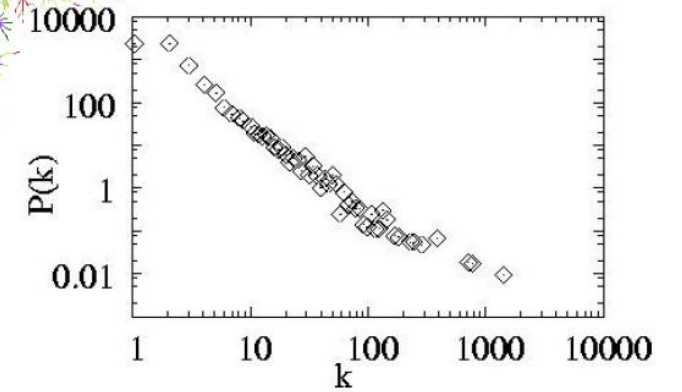
With exponent γ between **2.1 and 2.3**
Weakly depending on phenomena,
from World Wide Web,
to hardwired internet, to gene regulation.

Mini Tutorial: Which problems would there be in constructing a network with N nodes with exponent $\gamma = 1.8$?

Examples:



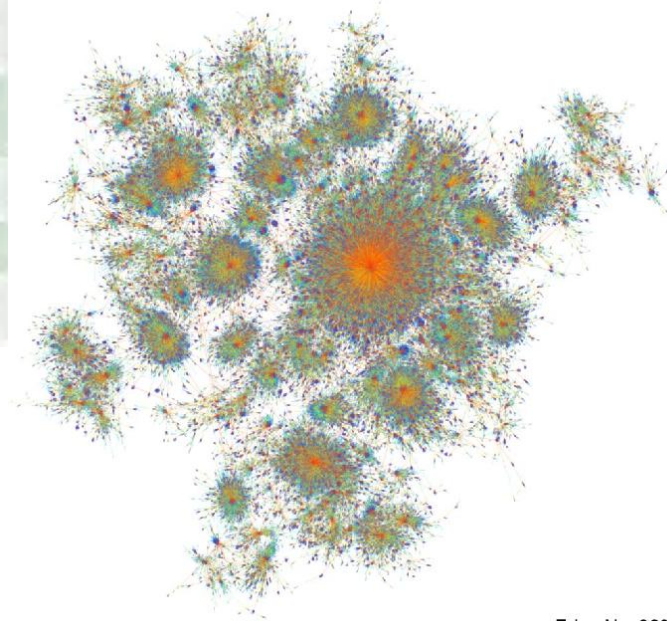
The Internet





Examples of Scale-Free Networks

NETWORK	NODES	LINKS
Cellular metabolism	Molecules involved in turning food for energy	Participation in the same biochemical reaction
Hollywood	Actors	Appearance in the same movie
Internet	Routers	Optical and other physical connections
Protein regulatory network	Proteins that help to regulate a cell's activities	Interactions among proteins
Research collaborations	Scientists	Co-authorship of papers
Sexual relationships	People	Sexual contact
World Wide Web	Web pages	URLs



Amplification in scale-free networks

$$\mathcal{A} = \frac{\langle k^2 \rangle}{\langle k \rangle} - 1 = \frac{\int_1^N \frac{k^2 dk}{k^\gamma}}{\int_1^N \frac{k dk}{k^\gamma}} - 1 \sim N^{3-\gamma}$$

For γ between 2 and 3, implying that $\mathcal{A} \rightarrow \text{infinity}$ as N increases

This reflects **the preferential spread to high-degree nodes**,
From where the rest of the network is reached.
(people with higher connectivity become infected first)

Vaccination (of a fraction f of people)

$$\mathcal{A} \rightarrow \mathcal{A}' = \mathcal{A} \cdot (1 - f)$$

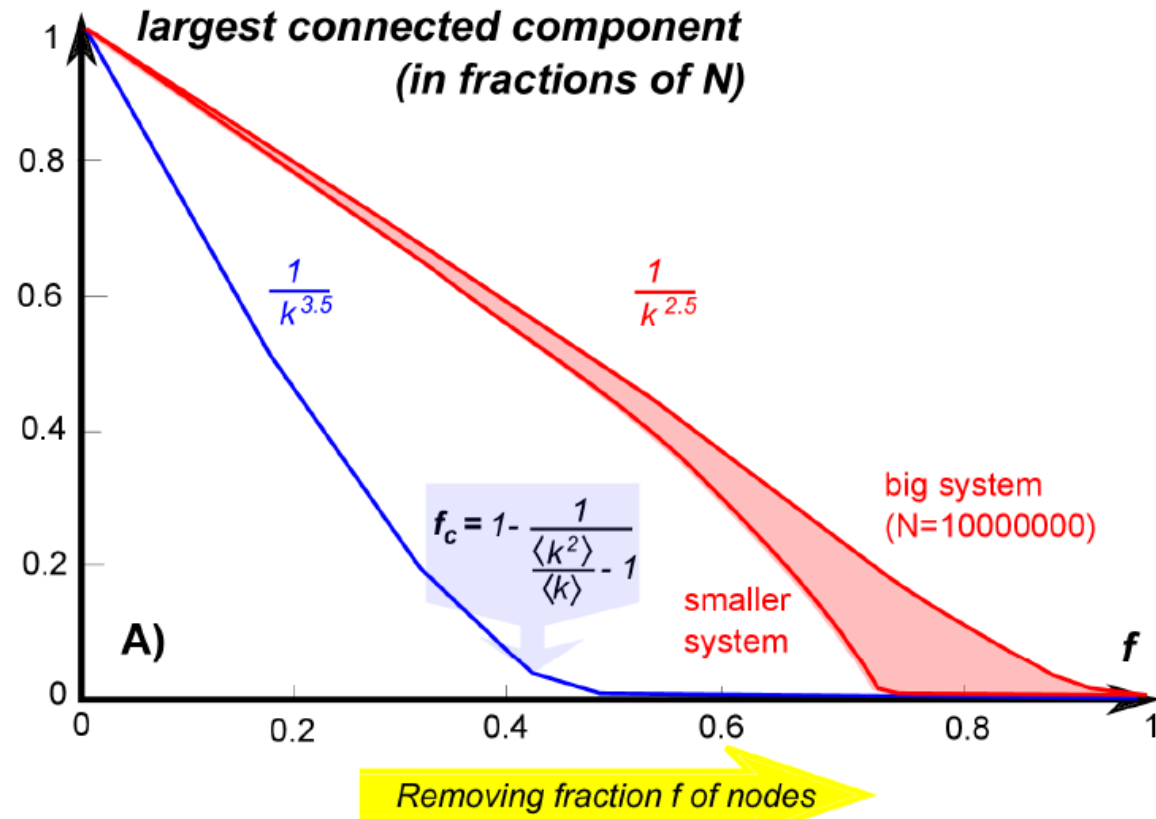
This comes about because each remaining node will lose each of its links with probability $f' = f$ ⁴⁾.

Disease will transmit when:

$$\mathcal{A} \cdot (1 - f) > 1 \quad \longrightarrow \quad (1 - f) > \frac{1}{\mathcal{A}} = \frac{1}{\langle k^2 \rangle / \langle k \rangle - 1}$$

$$\text{Vaccinate fraction: } f_c = 1 - \frac{1}{\langle k^2 \rangle / \langle k \rangle - 1} \longrightarrow 1, \text{ i.e. everybody}$$

Largest cluster as a function of vaccination fraction $f=1-p$, where p would be conducting nodes:



Tutorial:

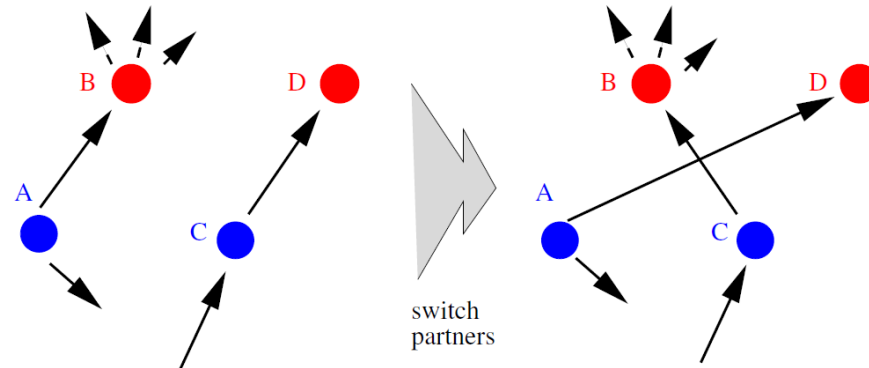
- Comment on a strategy where we vaccinate a random fraction of neighbors instead of a random fraction

... real world strategy against mpox ...
(using limited smallpox vaccine)

Detecting Patterns in Networks

(beyond the degree distribution)

- Randomization with preserved link number:



$$\text{Fraction}(\text{unchanged links}) = \left(1 - \frac{2}{L}\right)^t \approx e^{-2t/L}$$

Needed number of swaps: $e^{-2t/L} \sim 1/L \Rightarrow t \sim \frac{1}{2} L \cdot \ln(L)$

Quantification of pattern type Q:

$$Substance = Sub(pattern) = \frac{N(Q)}{\langle N_{random}(Q) \rangle}$$


Old fashioned concept

$$Significance = Z_{score} = \frac{N(Q) - \langle N_{random}(Q) \rangle}{\sigma_{random}(Q)}$$

$$\sigma_{random}^2(Q) = \langle N_{random}(Q)^2 \rangle - \langle N_{random}(Q) \rangle^2$$

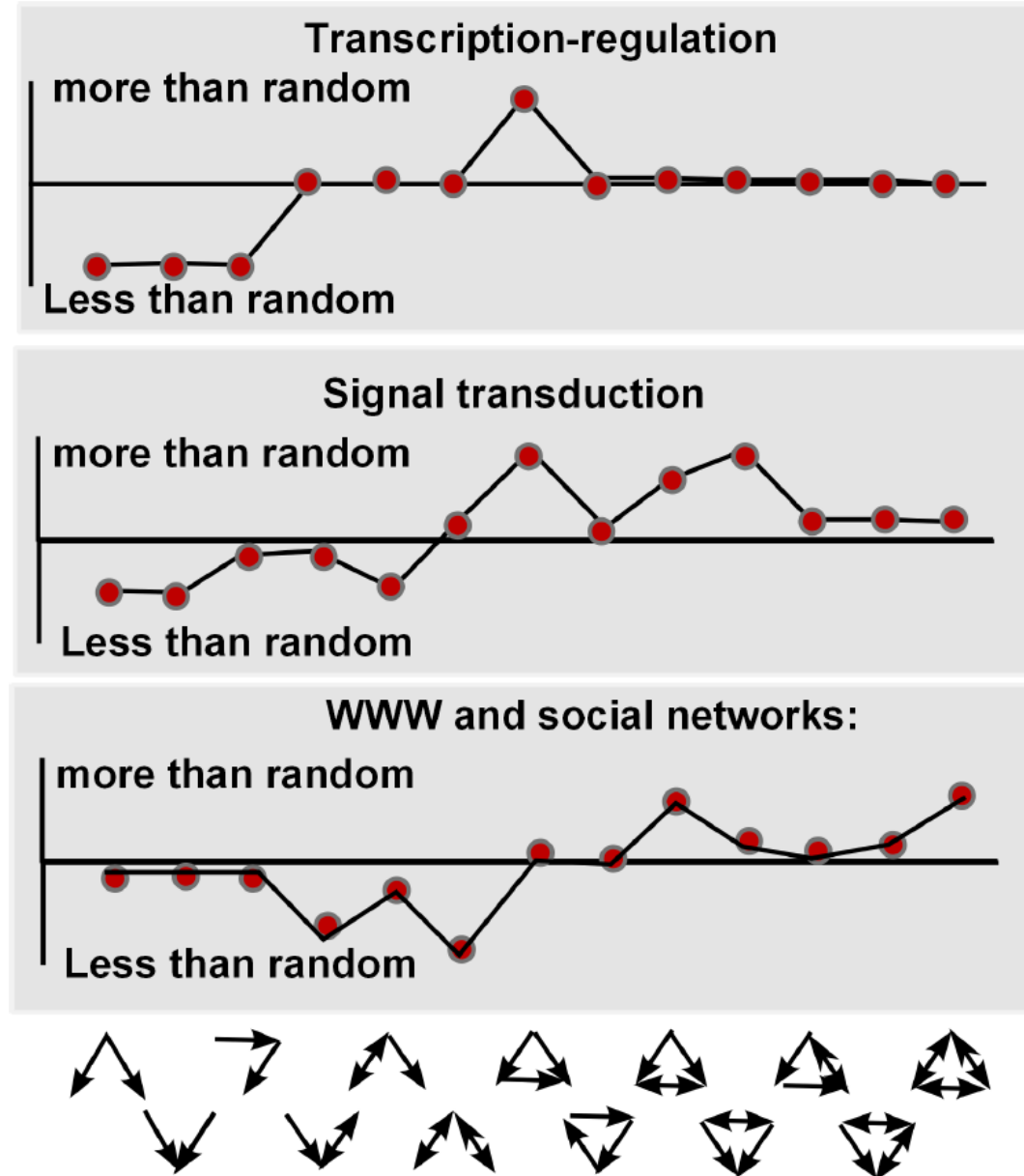
**Where N is the number of patterns of type Q in a network,
And average is over many random networks.**

Tutorial:

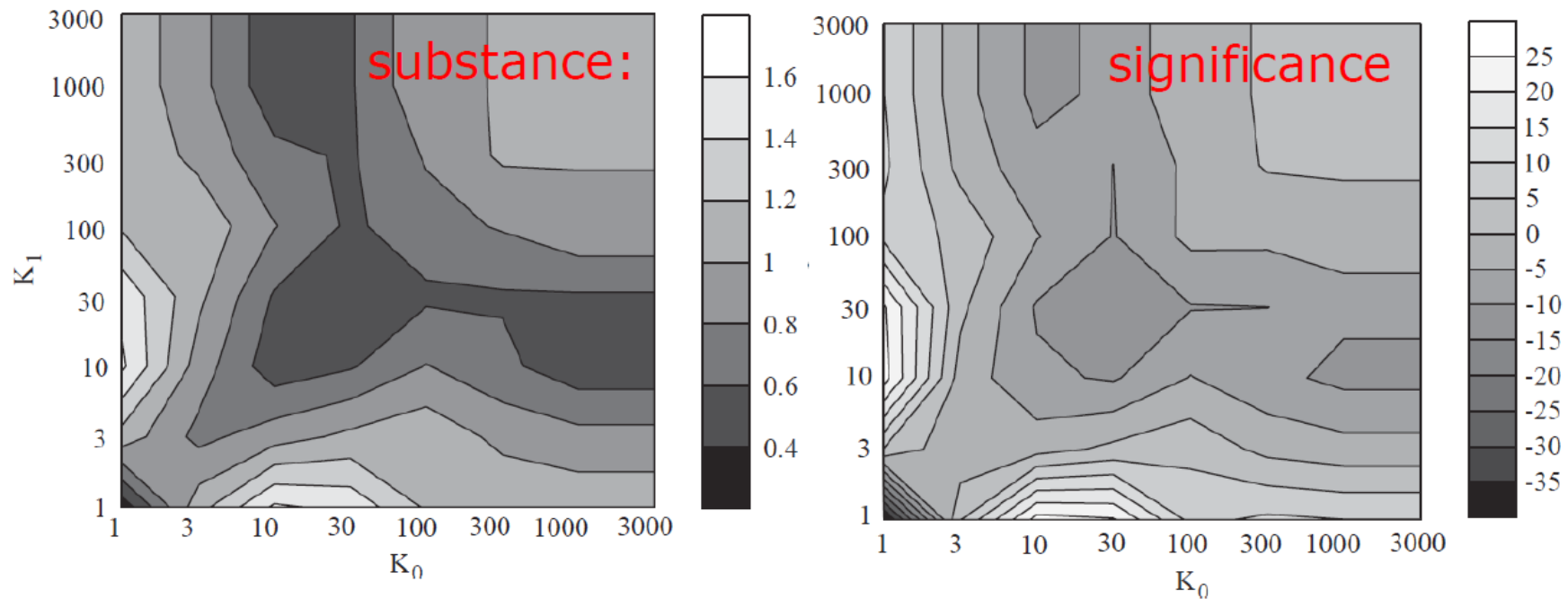
- Give some real-world examples where Sub is high and Z small
- Give some real-world examples where Sub is low but Z very high

Some abundant 3-node motifs:

Notice triangles often are favored,...
..but the type of triangles differs between
systems.

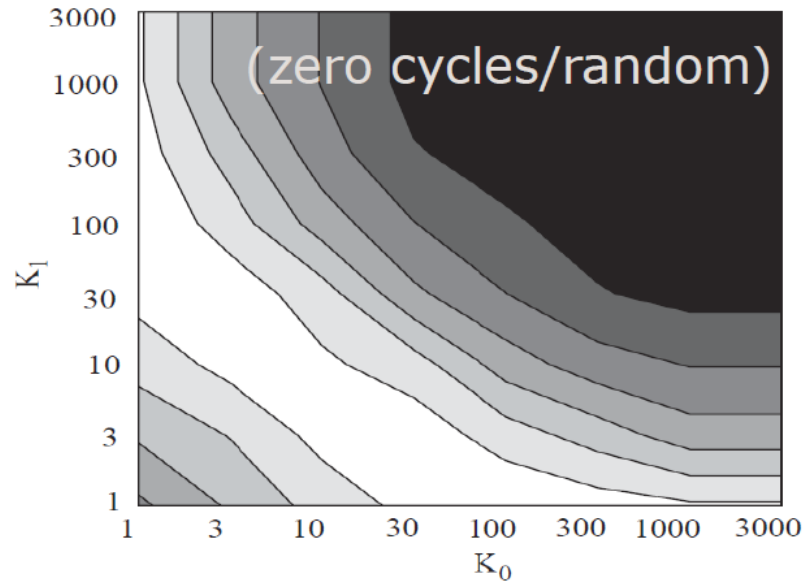


Connectivity patterns between nodes in the hardwired internet (servers):

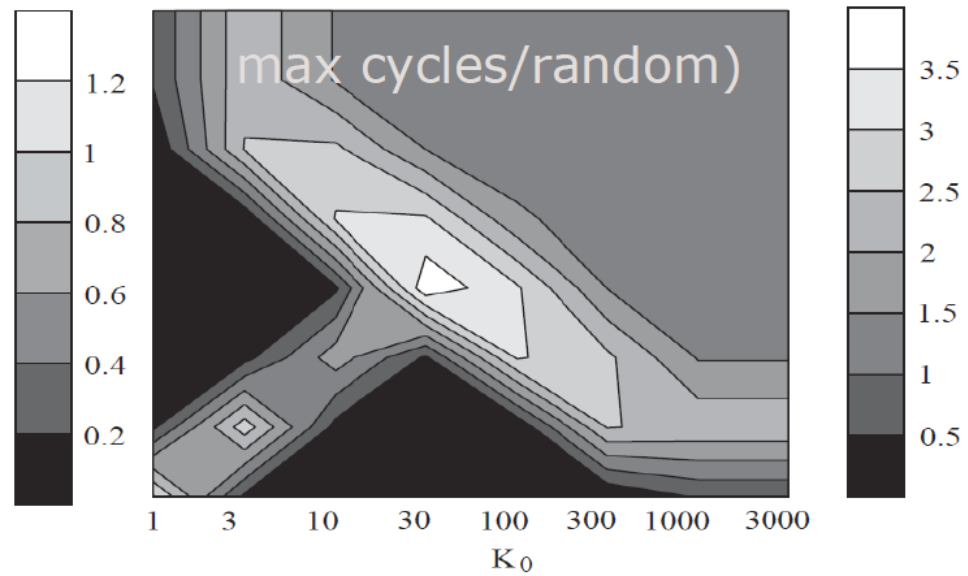


**Highly connected nodes (K about 30) connect to lower connected nodes ($K=1$ or 2),
 $K=30$ avoid connecting to other $K=30$ nodes**

Analyze **artificial** networks with the same k distribution (substance):



Avoid loops of length 3,
And the compare to random



Maximize the number of loops of length 3,
And then compare to random

Metropolis sampling while fixing some pattern

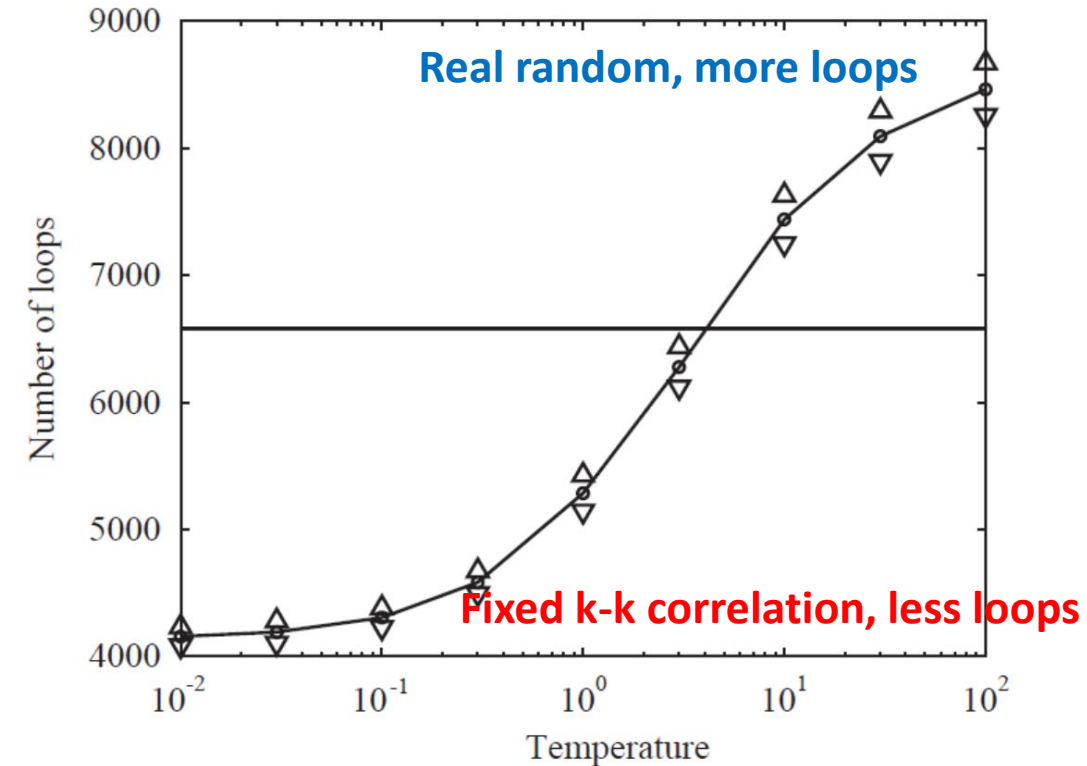
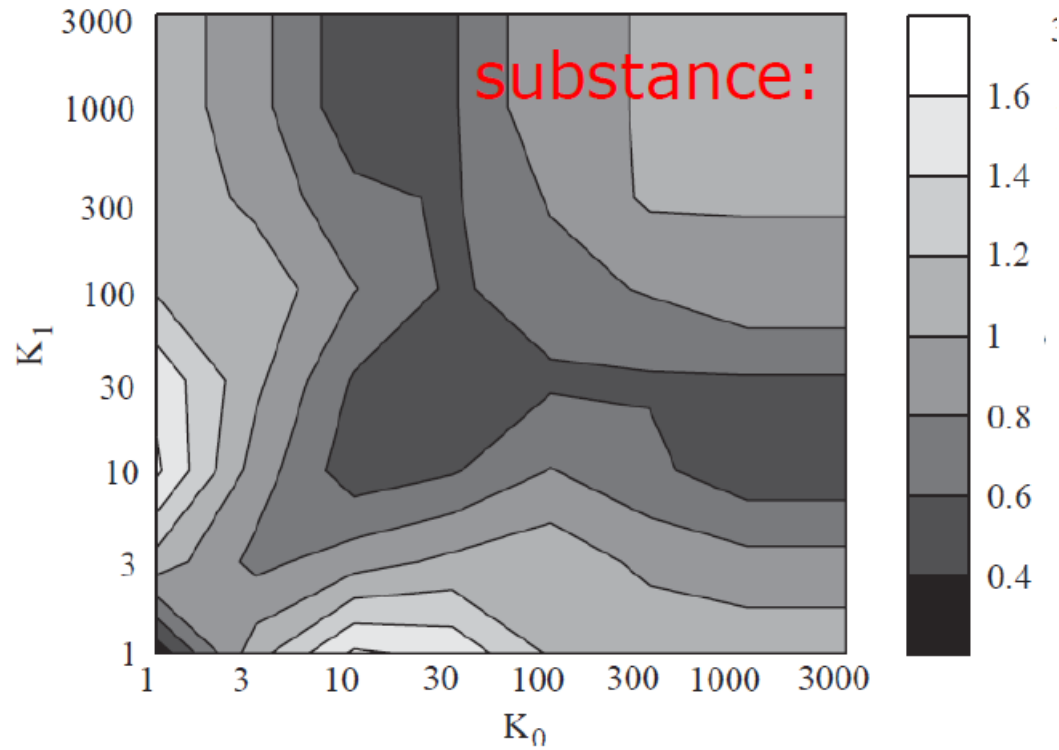
$$H = \sum_{K_0, K_1} (N(K_0, K_1) - N_{random}(K_0, K_1))^2 / N(K_0, K_1)$$

Attempt to make a swap=move:

- If $H(new) < H(old)$, move is accepted.
- If $H(new) > H(old)$: move is accepted with prob. $p = e^{-(H(new)-H(old))/T}$

Where T is some dummy temperature that can be changed:

Randomization (of the internet) while fixing correlation between neighbor k - k



Truly randomized ($T \gg 1$) have more loops,
while random with conserved k - k correlation
($T=0$) have fewer loops than real hardwired internet.

Overall lesson: $P(k) \rightarrow k$ - k correlation $\rightarrow$ loops (depend on $p(k)$ and k - k)

Recapitulation:

In network form:



In matrix form:

$$\mathbf{A} \mathbf{v}_1 = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$\begin{matrix} 1 & 2 & 3 \\ \text{to} & \text{to} & \text{to} \\ j & j & j \end{matrix}$

Virus
on networks:

$$n(\Delta) = \frac{1}{6} \cdot \text{trace}(\mathbf{A}^3)$$

→ measure of locality... how many of your friends that are friends

Random walk
on networks:

$$\frac{v_i^1}{k_i} = \frac{v_j^1}{k_j} = \text{constant}$$

→ measure of popularity, &
Measures of large distant clusters.

$$n(k) \propto \frac{1}{k^\gamma} \Rightarrow \mathcal{A} = \frac{\langle k^2 \rangle}{\langle k \rangle} - 1 = \frac{\int_1^N \frac{k^2 dk}{k^\gamma}}{\int_1^N \frac{k dk}{k^\gamma}} - 1 \sim N^{3-\gamma}$$

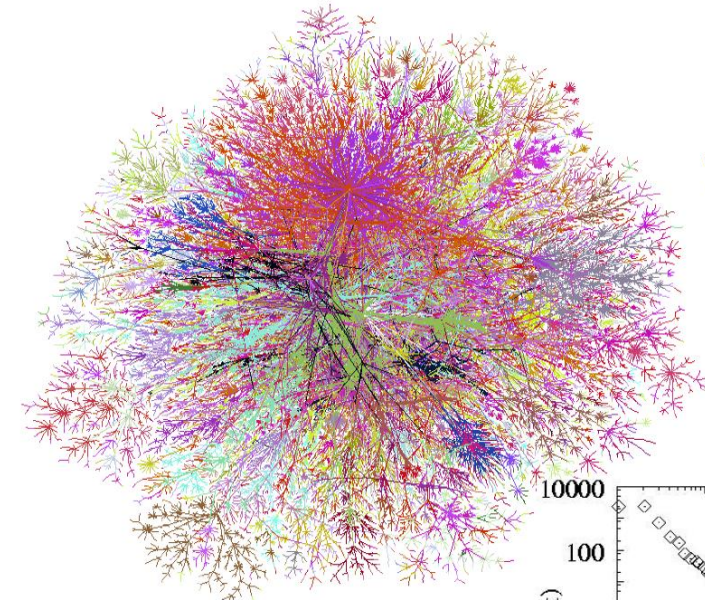
Recapitulation:

Scale-free networks:

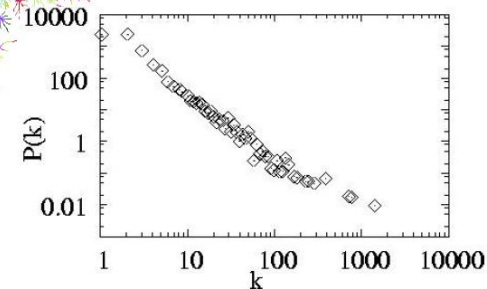
$$n(k) \propto \frac{1}{k^\gamma}$$

With exponent γ between
2.1 and 2.3

$$\mathcal{A} = \frac{\langle k^2 \rangle}{\langle k \rangle} - 1 = \frac{\int_1^N \frac{k^2 dk}{k^\gamma}}{\int_1^N \frac{k dk}{k^\gamma}} - 1 \sim N^{3-\gamma}$$



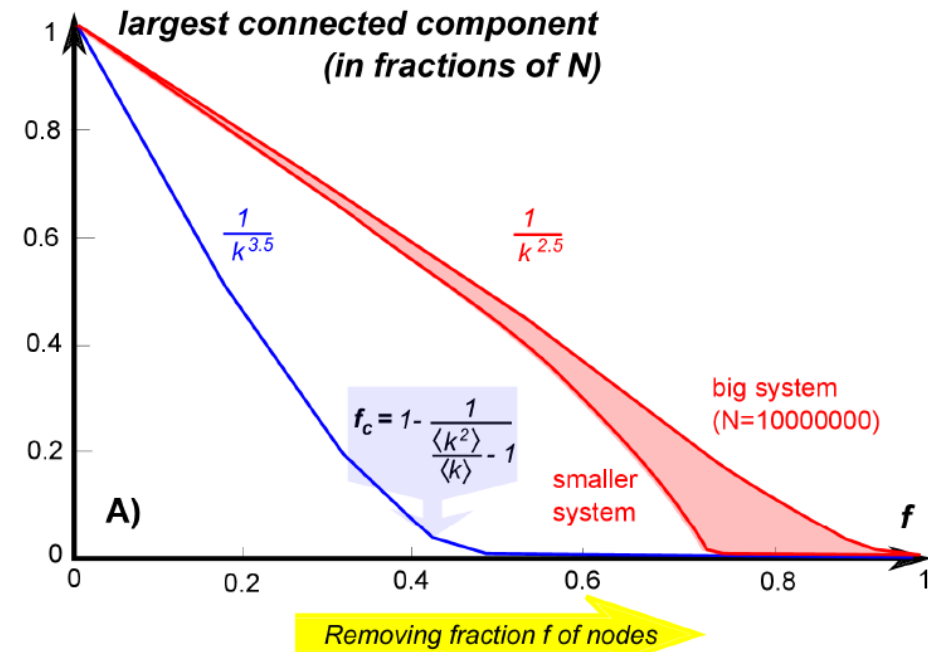
The Internet



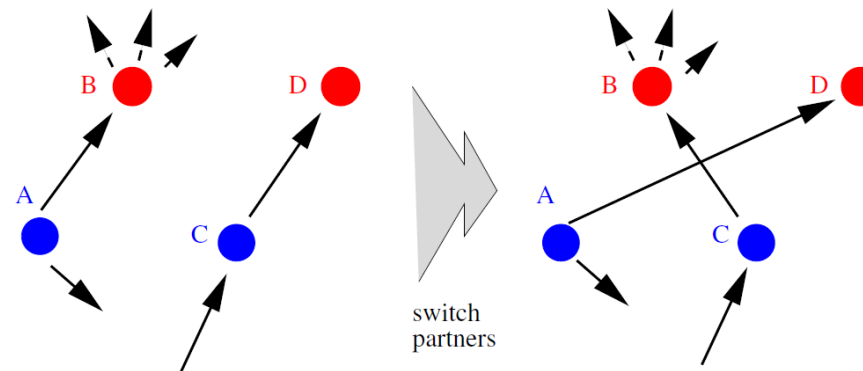
Recapitulation:

Vaccinate fraction:

$$f_c = 1 - \frac{1}{\langle k^2 \rangle / \langle k \rangle - 1}$$



Randomized null model:



$$Substance = Sub(pattern) = \frac{N(Q)}{\langle N_{random}(Q) \rangle}$$

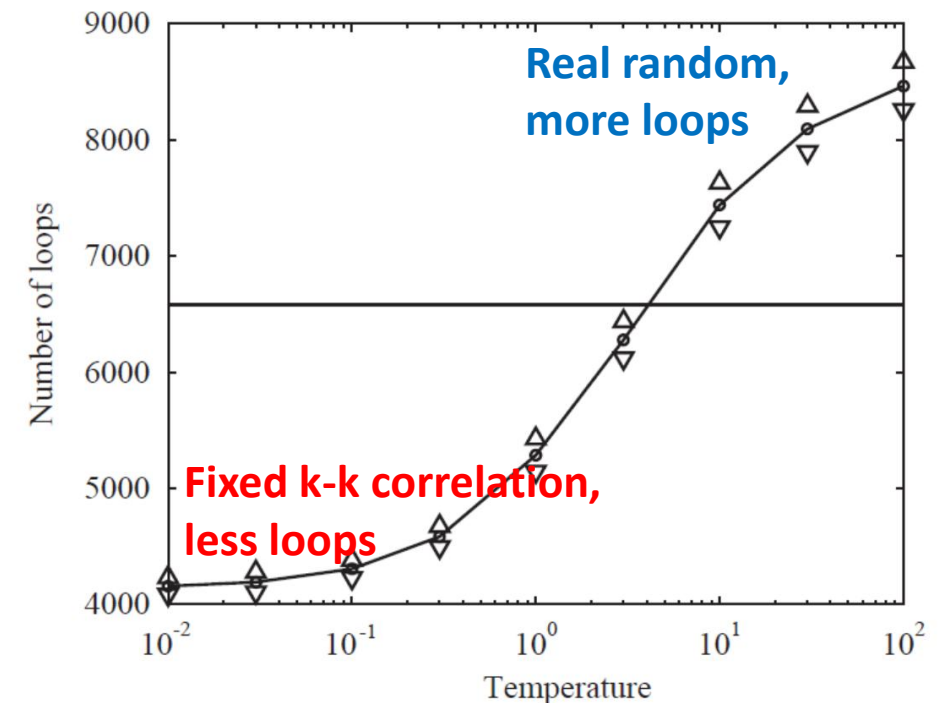
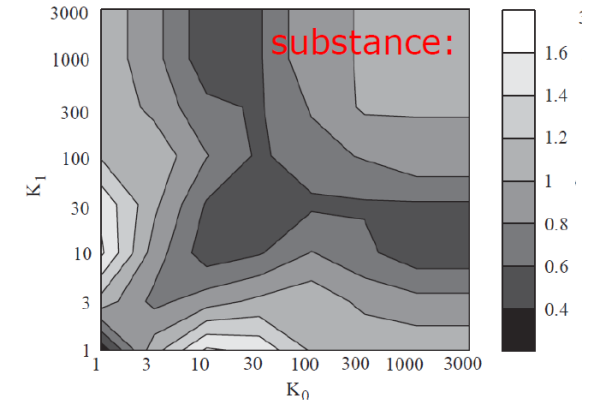
Recapitulation: Metropolis sampling of networks with a fixed pattern

$$H = \sum_{K_0, K_1} (N(K_0, K_1) - N_{random}(K_0, K_1))^2 / N(K_0, K_1)$$

Attempt to make a swap=move:

- If $H(new) < H(old)$, move is accepted.
- If $H(new) > H(old)$: move is accepted with prob. $p = e^{-(H(new)-H(old))/T}$

Where T is some dummy temperature
that can be changed →



How to make a random scale-free network with arbitrary exponent:

For each node assign $k_i = K$ from $\frac{\int_K^N dk/k^\gamma}{\int_1^N dk/k^\gamma} = r$ With r random $[0,1]$

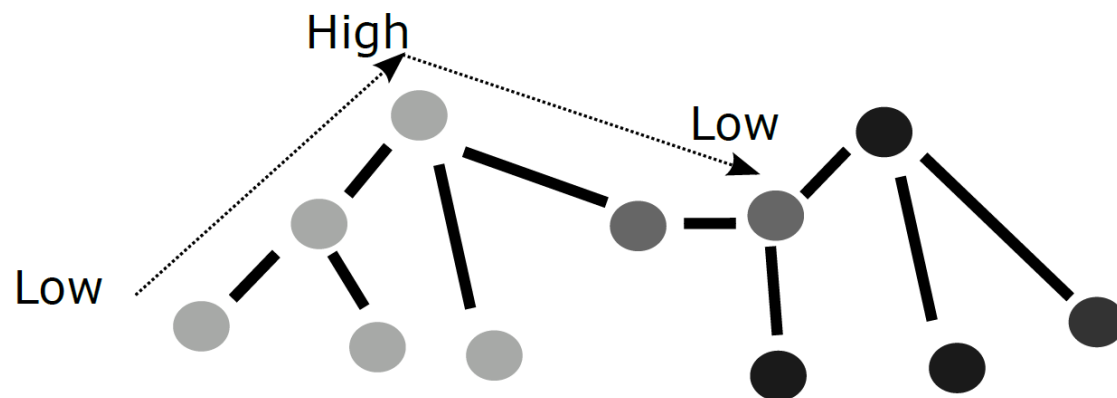
Lists nodes from high k and downwards, hierarchically.

Assign links starting from the highest node from top to bottom
→ highly assortative network

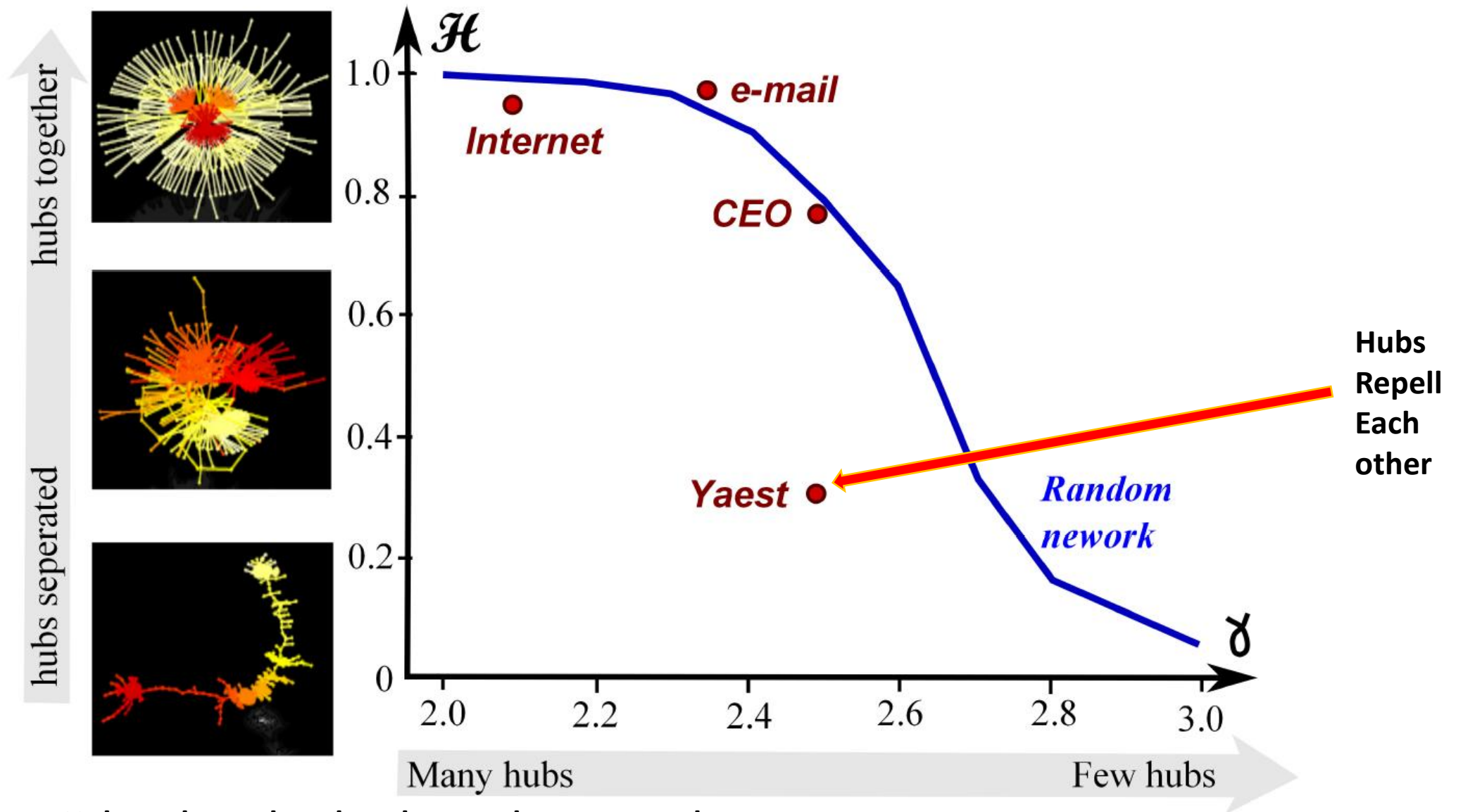
Randomize with edge swapping, of order $L \ln(L)$ swaps, where $L = (\sum_i k_i)/2$.

Tutorial: Problems if exponent $\gamma < 2$, WHY?

Hierarchy measure of networks:



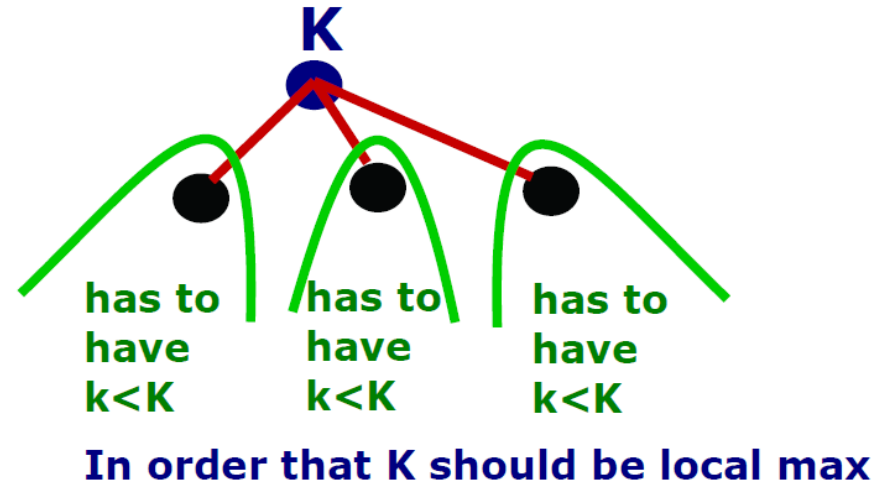
$$\mathcal{H} = \frac{\sum_{ij} (\text{shortest path between } i \text{ and } j \text{ hierarchical})}{N(N-1)/2}$$



Hubs so large that they have to be connected,
 i.e. two nodes i, j each connectivity 1000 in a $N=10,000$ network, with $\sum k=20,000$
 Each of the links from i have probability $1000/20000=0.05$ to connect to j
 In average this would give 50 links between i and j , where only one is allowed. **Difficult to avoid!**

Some intuition:

$$\begin{aligned}
 P(K \text{ node is local top}) &= \left(1 - \frac{\int_K^N k^{1-\gamma} dk}{\int_1^N k^{1-\gamma} dk} \right)^K \\
 &\sim (1 - K^{2-\gamma})^K \\
 &\sim \exp(-K^{3-\gamma}),
 \end{aligned}$$



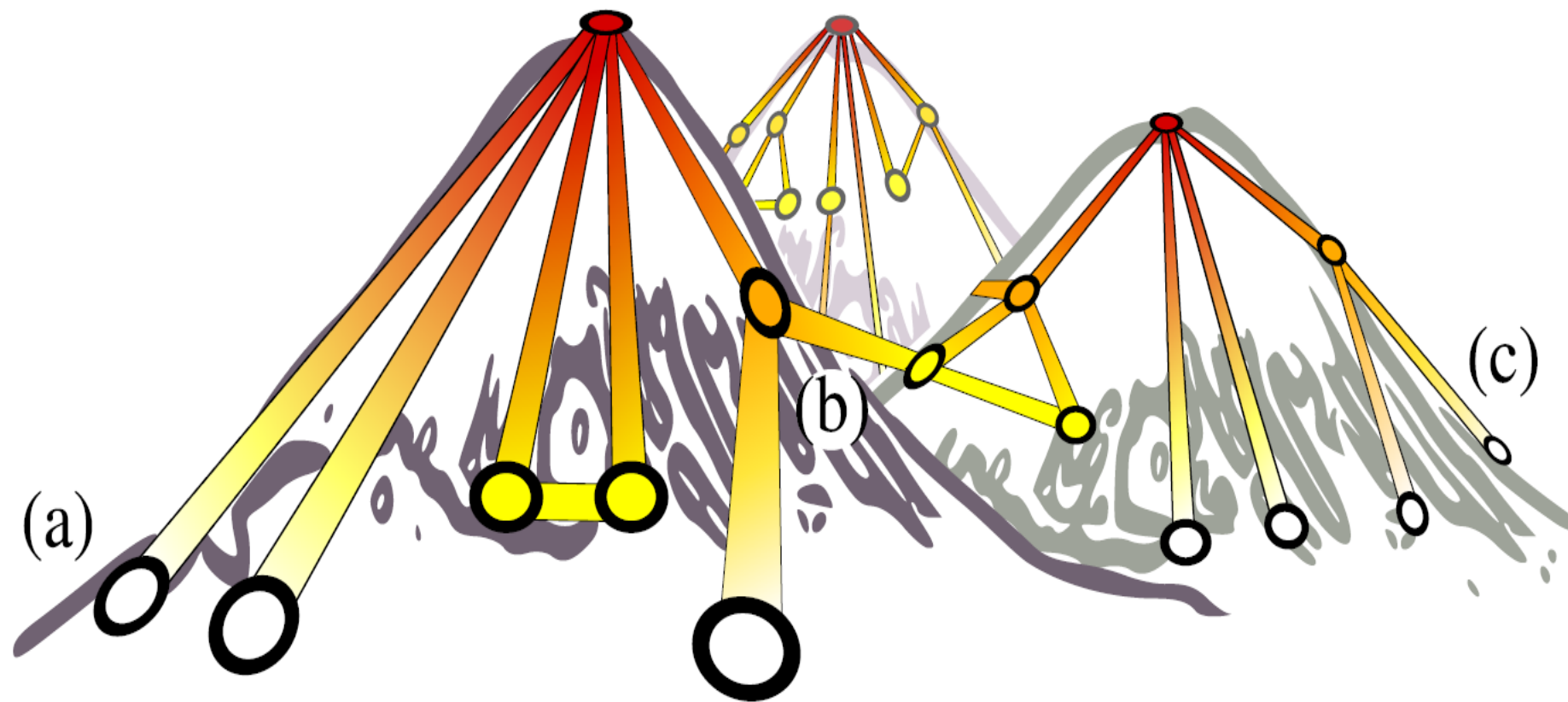
where we use that, when $\gamma > 2$, the integral is dominated by its lower boundary. Subsequently, we use that $1 - K^{2-\gamma} \sim \exp(-K^{2-\gamma})$. Thereby $P(K \text{ node is local top})$ becomes large for $\gamma \sim 3$, even when K is rather small.

$\gamma \sim 3$ the probability to be top becomes large $\rightarrow$ anti-hierarchical

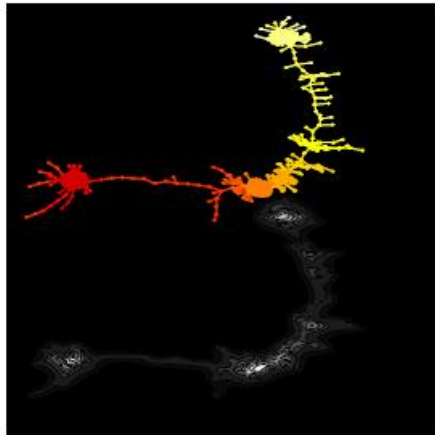
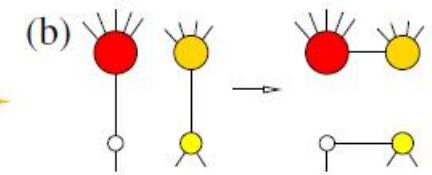
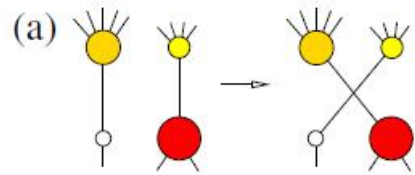
~ 2 no reasonable-sized k can be a top $\rightarrow$

only one top in the whole network $\rightarrow$ hierarchical

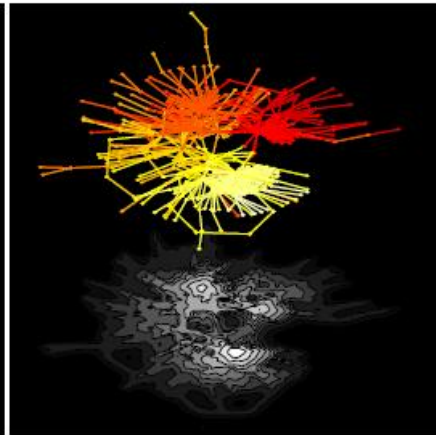
A non-hierarchical network, with many k-tops and k-valleys



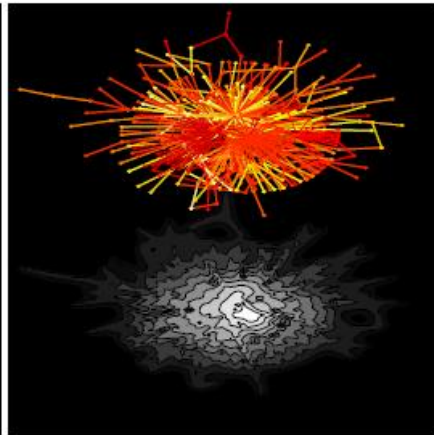
Reorganising a $1/k^\gamma$ between extreme topologies:



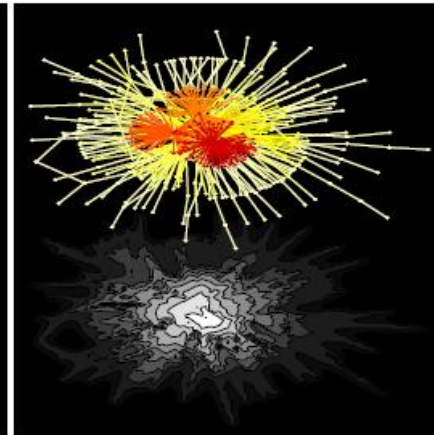
(c) Random rank, $\varepsilon = 0$
 $\mathcal{F} = 0.13(5)$



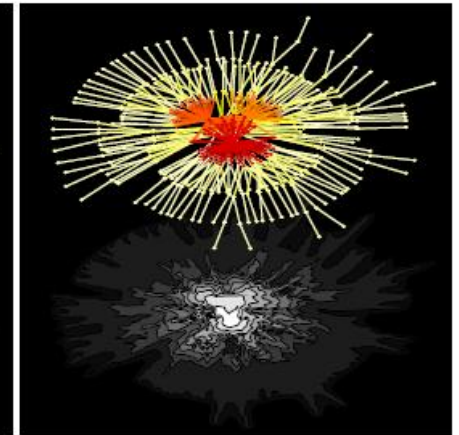
(d) Random rank, $\varepsilon = 0.1$
 $\mathcal{F} = 0.51(5)$



(e) Random, $\varepsilon = 1$
 $\mathcal{F} = 0.83(5)$



(f) Degree rank, $\varepsilon = 0.5$
 $\mathcal{F} = 0.96(3)$



(g) Degree rank, $\varepsilon = 0$
 $\mathcal{F} = 1.00(0)$

Assign random number r_i to each node i and swap to minimize distances of numbers between neighbours

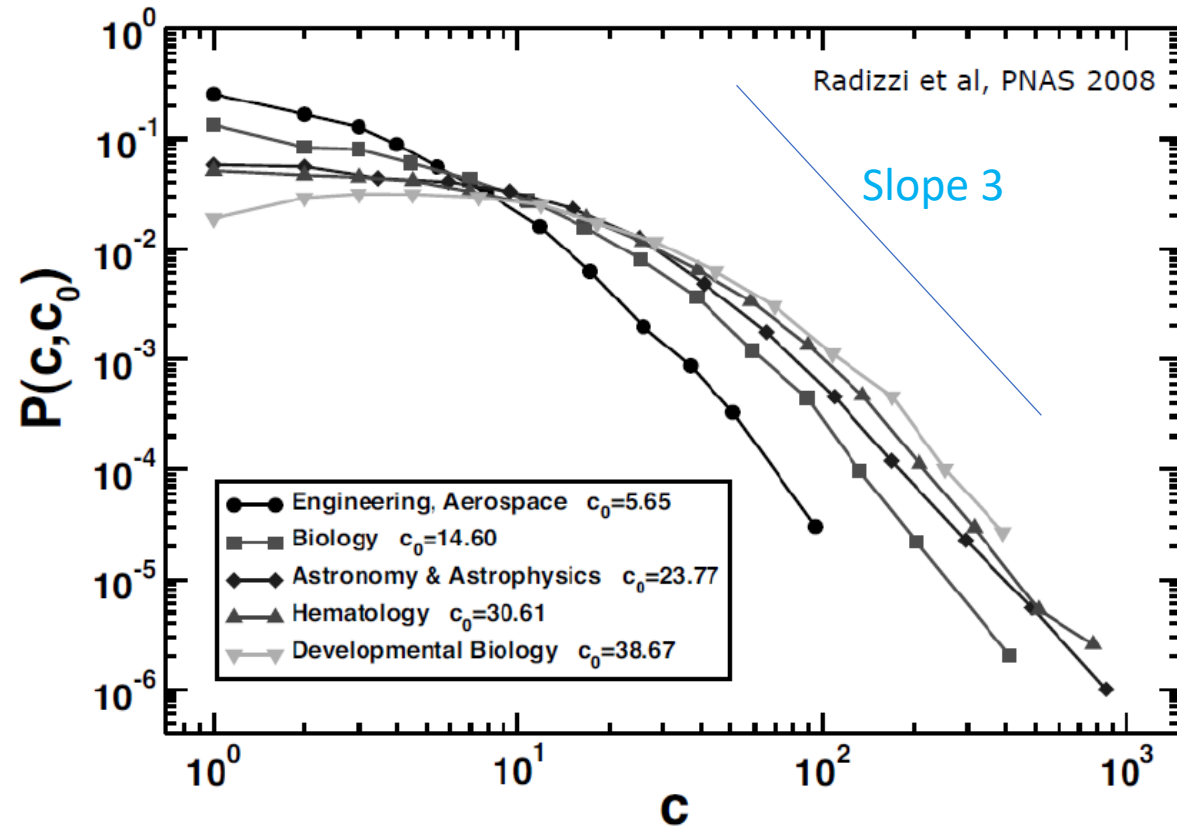
Epsilon:
Likelihood
Of doing random swap

Assign number $=ki$ to each node i and swap to minimize distances of numbers between neighbours

How does a Scale-free network emerge?

- Simple answer: **we have no clue**
- Complex answer: **There are many models which are all wrong**
- → will explore two such wrong models,
Also because they are right for some other phenomena

Preferential attachment/Cumulative advantage



Inspiration from citation networks, somewhat scale-free
with “exponent” close(?) to 3.. Or maybe more log-normal?

Minimal information → Preferential attachment (attachment along random walk on a network):

$$\frac{v_i^1}{k_i} = \frac{v_j^1}{k_j} = \textit{constant}$$

... don't bother to read articles, just follow jumps guided by reference lists...

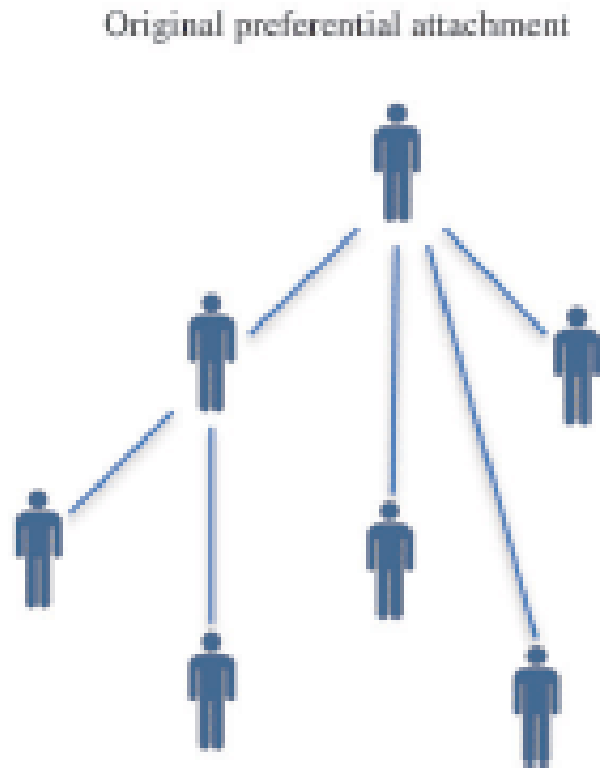
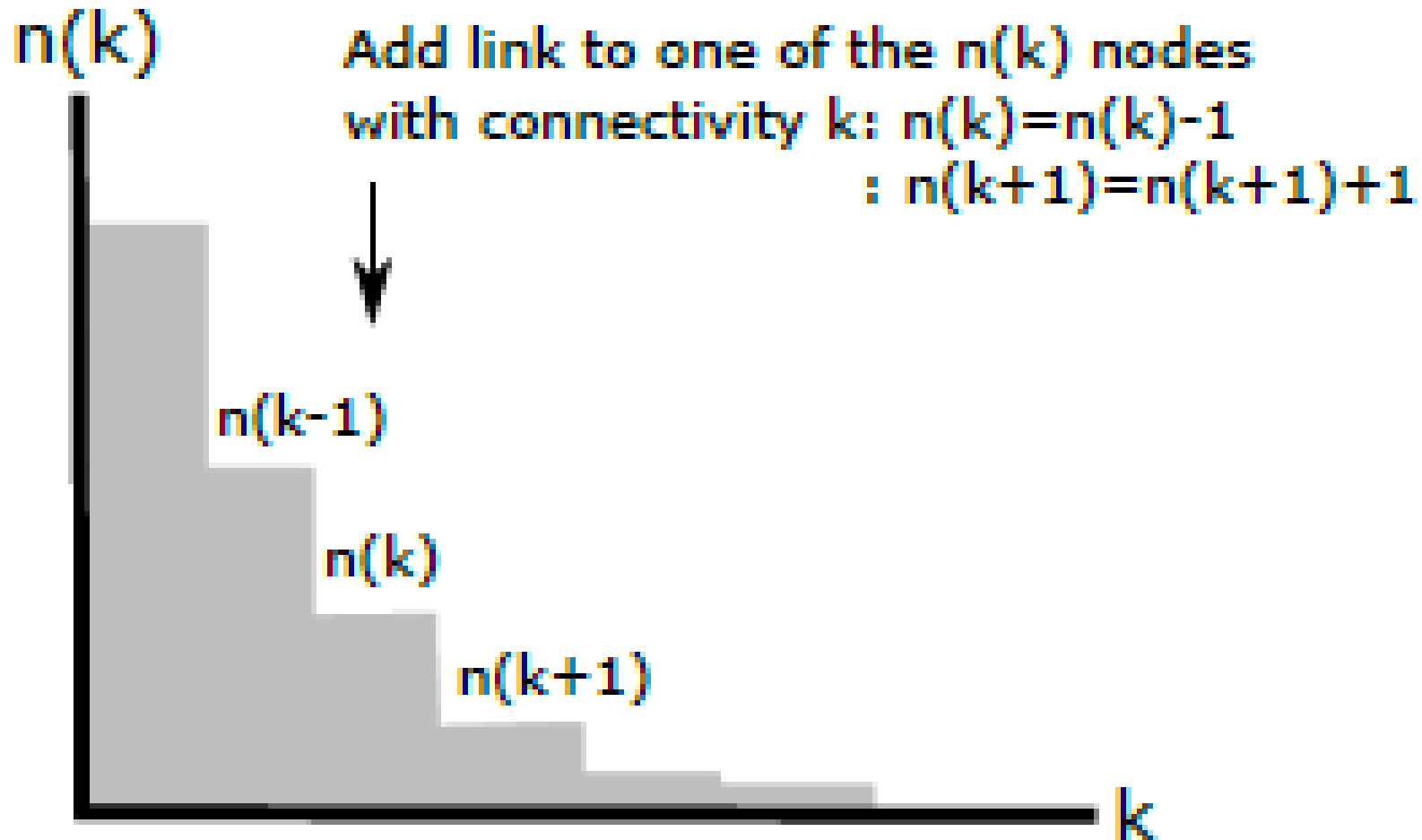


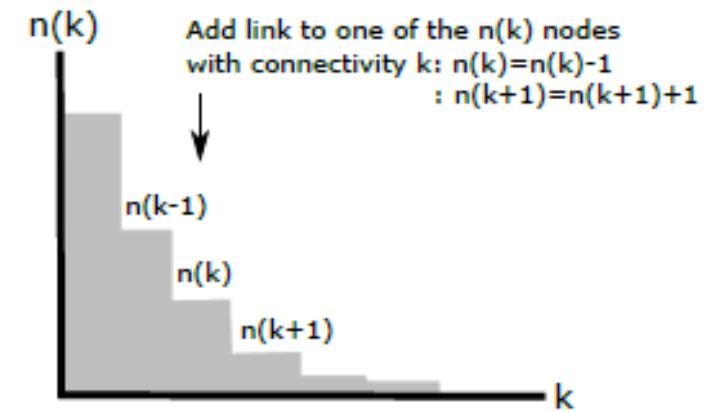
Figure 2: Network structure generation (Original preferential attachment vs Modified preferential attachment).

Consider the distribution of the number of nodes $n(k)$ with connectivity k :



Add to what already have:...

each new links comes from a process where one follow links and attach to nodes along the random walk path



$n(k, t)$ be the number of nodes with connectivity k at time t .

$$n(k, t+1) - n(k, t) = \frac{(k-1) \cdot n(k-1, t) - k \cdot n(k, t)}{\sum_{k'} k' n(k')} \quad \text{for } k > 1$$

$$\Rightarrow \frac{dn(k, t)}{dt} = -\frac{1}{\sum k' n(k')} \frac{d}{dk} (k \cdot n(k, t)) \quad \text{Use:} \quad \sum_k k n(k, t) = 2t$$

$$\Rightarrow \frac{dn}{dt} = -\frac{1}{2t} \frac{d(k \cdot n)}{dk} \quad \text{Ansatz:} \quad n(k, t) = f(t) \cdot N(k)$$

$$2t \frac{f'(t)}{f(t)} = -\frac{1}{N} \frac{d(kN)}{dk}$$

Preferential attachment: → too steep exponent

$$2t \frac{f'(t)}{f(t)} = -\frac{1}{N} \frac{d(kN)}{dk}$$

where $f(t) \propto t$ since $f(t) = 2t / \sum_k kN(k)$

Inserting $f'/f = 1/t$

$$-1 - \frac{d \ln(N)}{d \ln(k)} = 2 \quad \Rightarrow \quad n(k) \propto \frac{1}{k^3}$$

Tutorials:

Mini Tutorial: What happens with

$$n(k, t + 1) - n(k, t) = \frac{(k - 1) \cdot n(k - 1, t) - k \cdot n(k, t)}{\sum_{k'} k' n(k')} \quad \text{and} \quad \sum_k k n(k, t) = 2t$$

if one assign two links (instead of one) to each new node added to the system?

What if one just add links preferentially in both ends?

Mini Tutorial: Argue why rich-get-richer dynamics could be considered for sizes of families of related species on ecological scale (Youle's law).

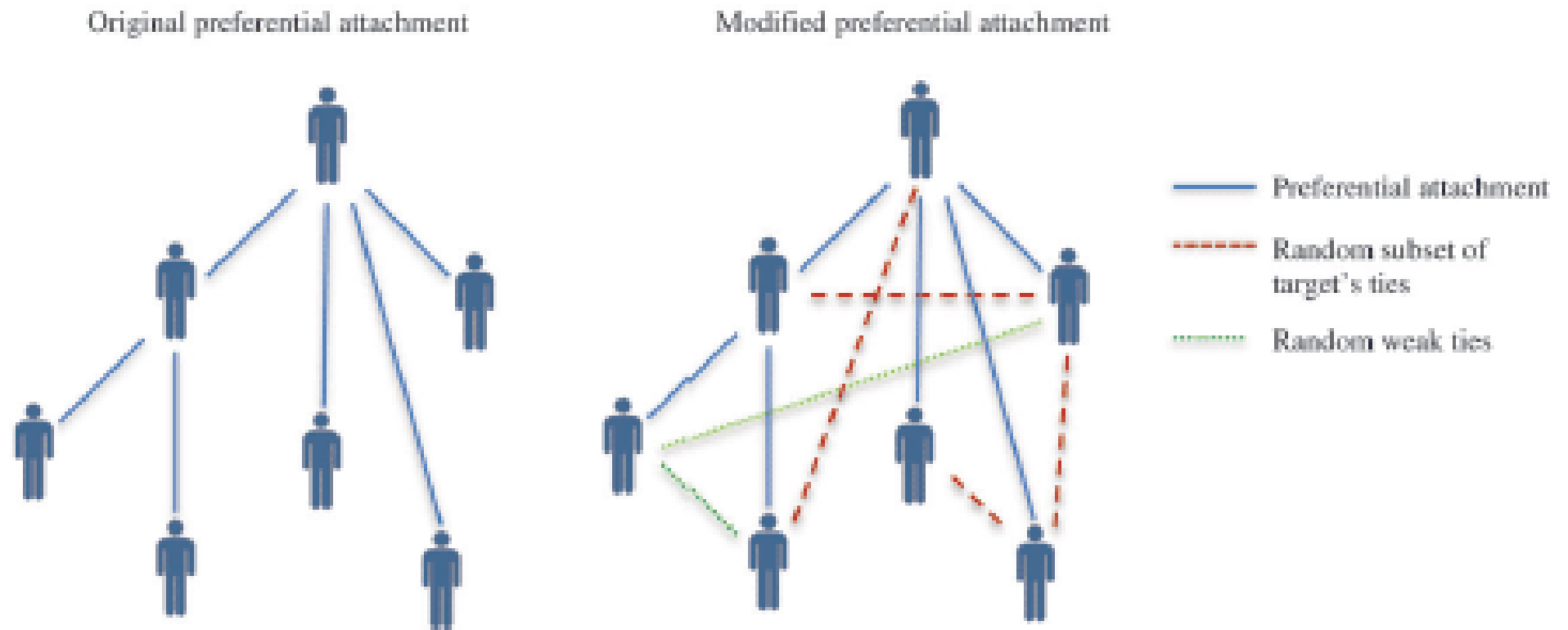
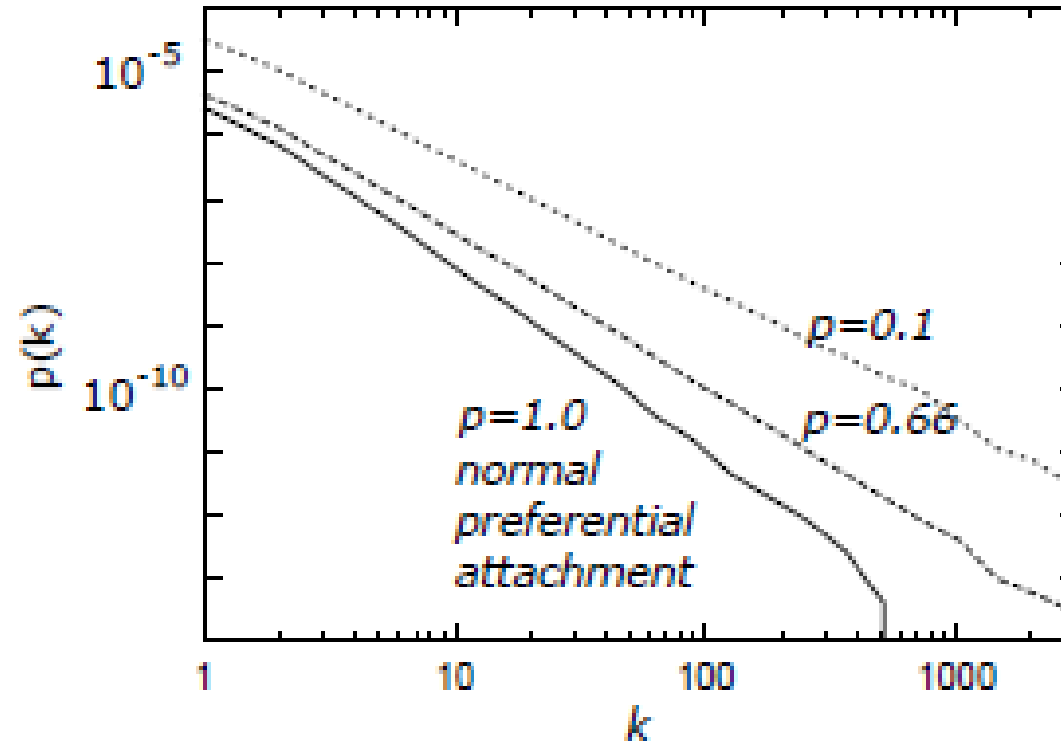


Figure 2: Network structure generation (Original preferential attachment vs Modified preferential attachment).

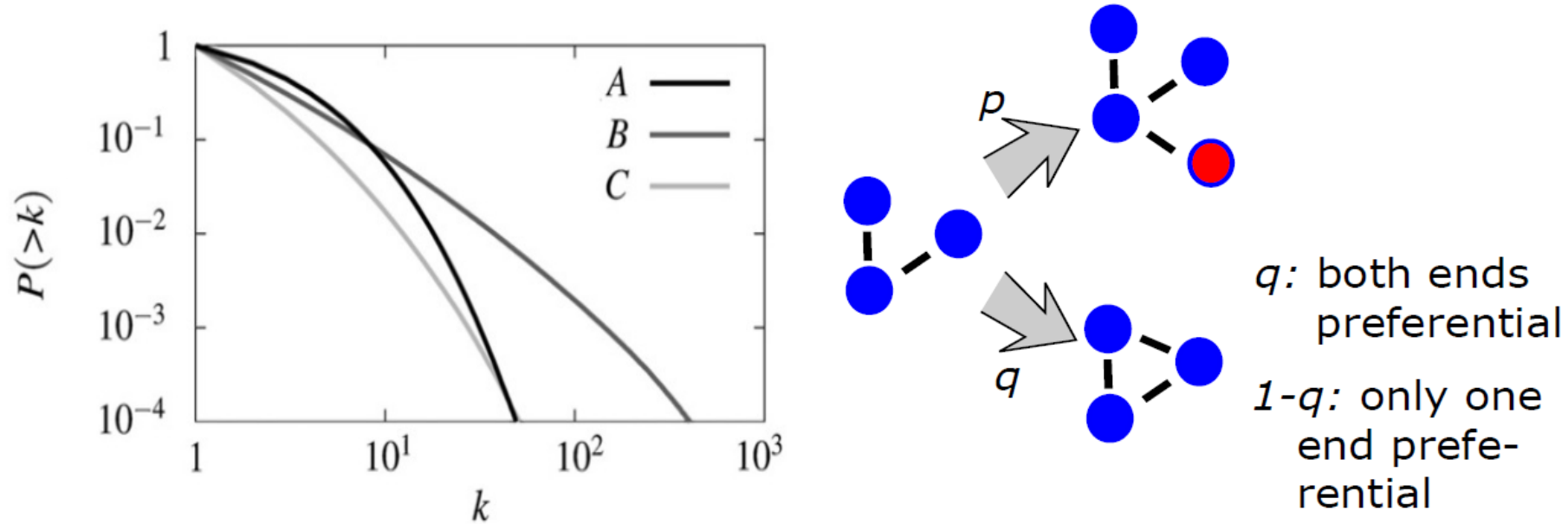
$1-p$ =probability to add link preferentially in both ends
 p =add node + link all preferentially.



→ But why? How could one be preferential on both ends?
where does information come from?

Preferential attachment with random killing of nodes: (when system size reaches 1000)

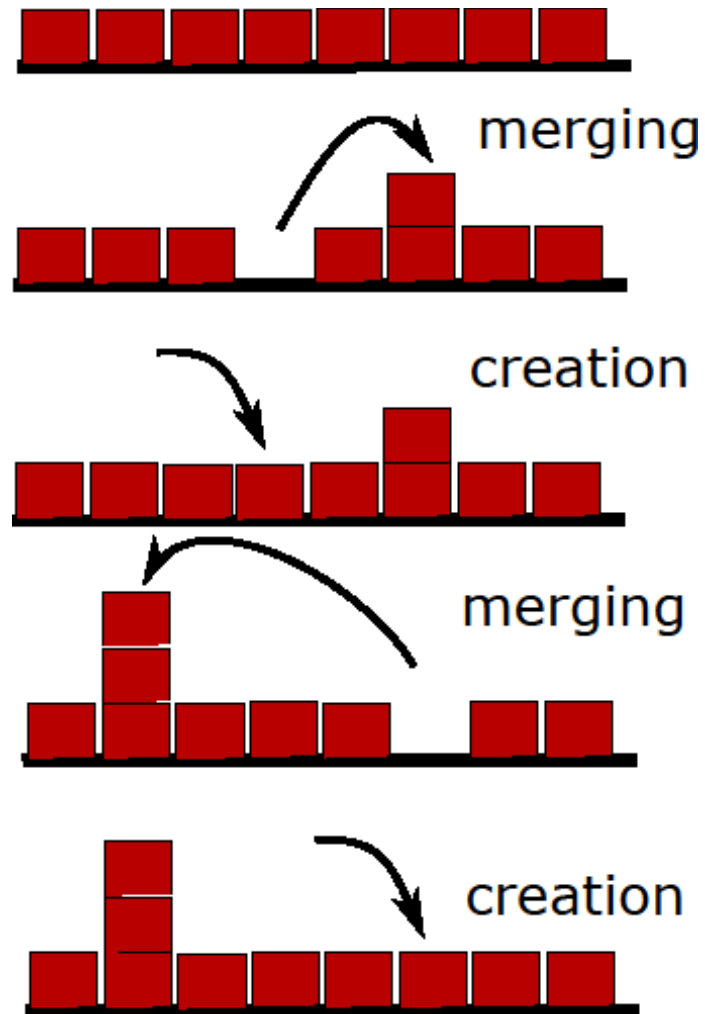
Preferential growth with random elimination ($N=1000$):



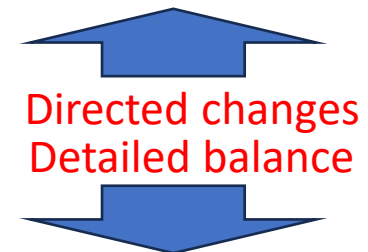
→ Why not? How could one it ever work
in some kind of steady state...

Merging and creation

Consider a set of N numbers, s_i , $i = 1, 2, \dots, N$. Initially, all these numbers are set equal to unity. At each time step select two of these numbers, say s_i and s_j , at random and add them, yielding a new number $s_i(new)$: $s_i(new) = s_i + s_j$. Then reset the number at position j to unity: $s_j = 1$.



Mini Tutorial: What would happen if one only merged, and did not inject, any new particles in the above merging and creation model?



Consider 100 numbers along a line, all set all numbers =1 initially. Take 2 random numbers A and B randomly. Add the two together, and partition them to A' and B' such that $A' + B' = A + B$. Only partition in natural whole numbers, 0,1,2,... Iterate the above 1000000 times and plot the histogram of the obtained values of individual numbers.

Probability to have size s :

$$p(s, t + 1) - p(s, t) = \sum_{u=1}^{s-1} p(s - u, t) \cdot p(u, t) - 2 \cdot p(s, t)$$

Gain of merging
from smaller
members

Loss because
a given member
could be selected
Twice.

$$p(s, t + 1) - p(s, t) = 0$$

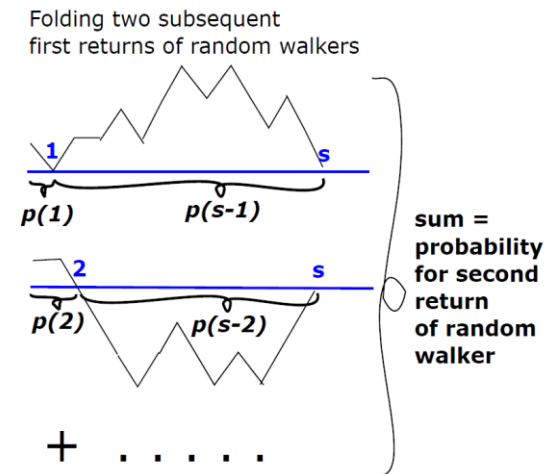
$$\Rightarrow \sum_{u=1}^{s-1} p(s - u) \cdot p(u) = 2 \cdot p(s)$$

Solved by analogy.

If $p(s)$ = first return of random walker along a time variable s

~ second return of random walker.

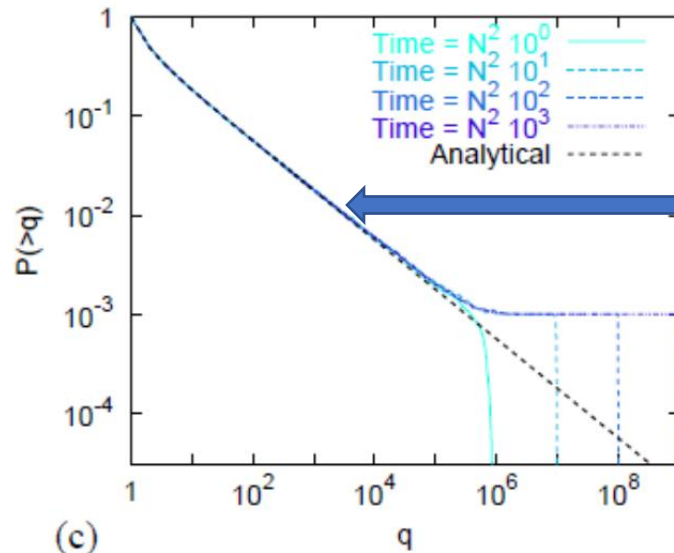
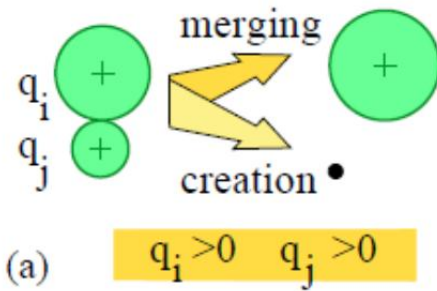
The axis that changes with the walk is not important, it is only a mathematical trick.



$$p(s) \propto \frac{1}{s^{3/2}}$$

The recurrent 3/2 exponent:

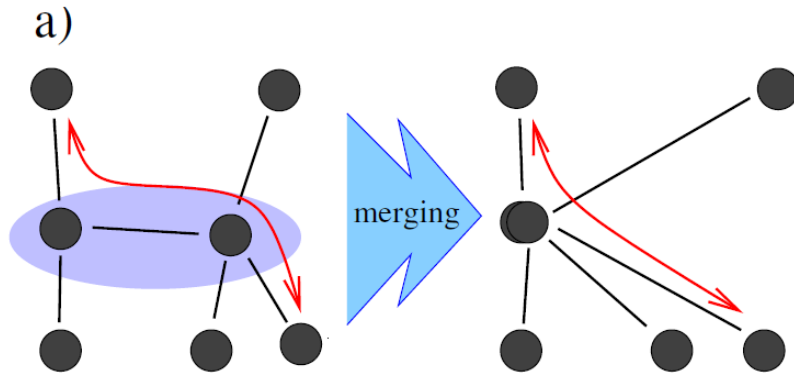
$$\left(\sum_u u^{-3/2} (s - u)^{-3/2} \approx 5.22 \cdot s^{-3/2}\right.$$



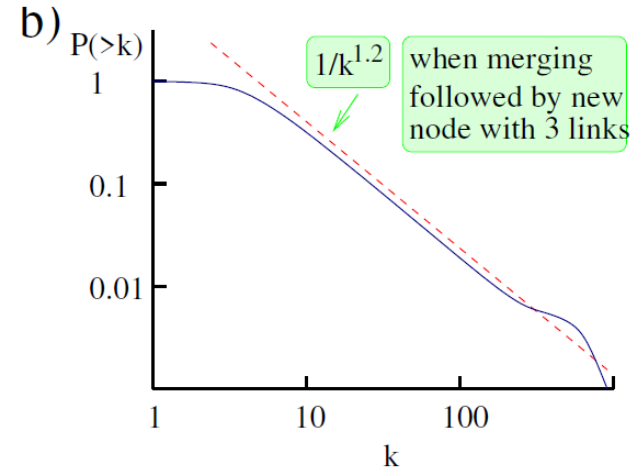
$$p(s) \propto \frac{1}{s^{3/2}}$$

This line reflect the linear growth of the largest cluster
 → going towards a sun-.

Now to merge nodes in networks:



Merging: Shortening signaling pathways



- Selecting a random node and one of its neighbors and merge them into one node.
- Adds one new node to the network and link it to a few randomly selected nodes.



Connectivity $p(k)=1/k^{2.2}$,.... **Close to typical network exponents**

Why the steeper exponent:

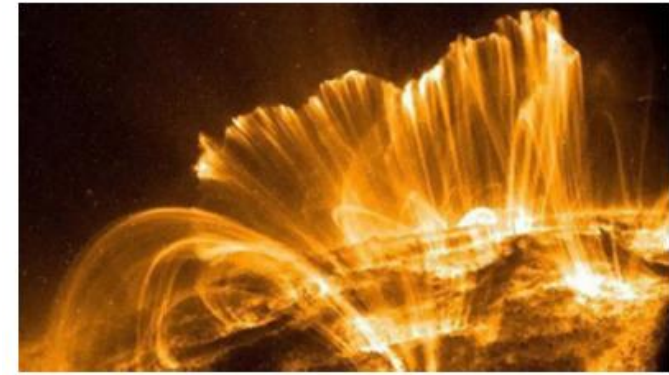
(from average mass conservation in merging-creation)

select two numbers of which one should be larger equal 1. These two number are merged but with removal of 1 from the aggregate: $(s_i, s_j) \rightarrow (s_i + s_j - 1, r)$. The injection r now is some random injection with mean $\langle r \rangle = 1$.

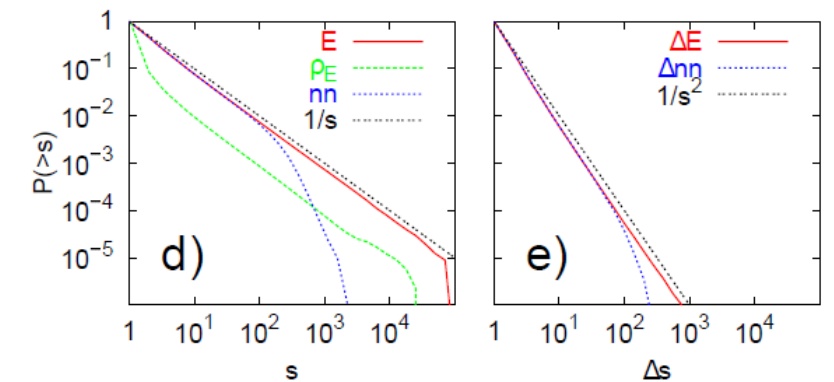
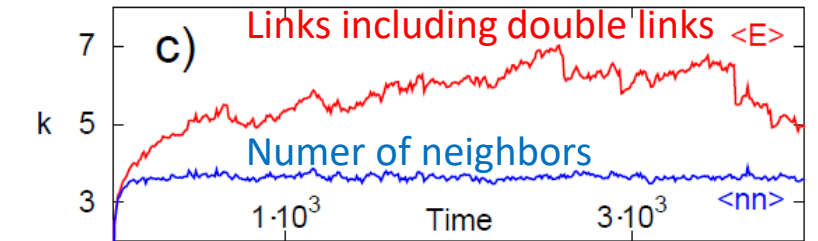
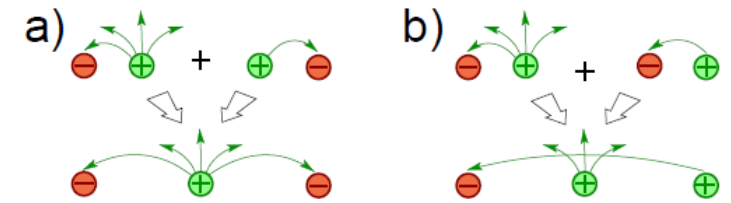
$$p(s) \propto 1/s^{5/2}$$

...much steeper than number merging with on going injection of mass

Merging bi-partite networks:



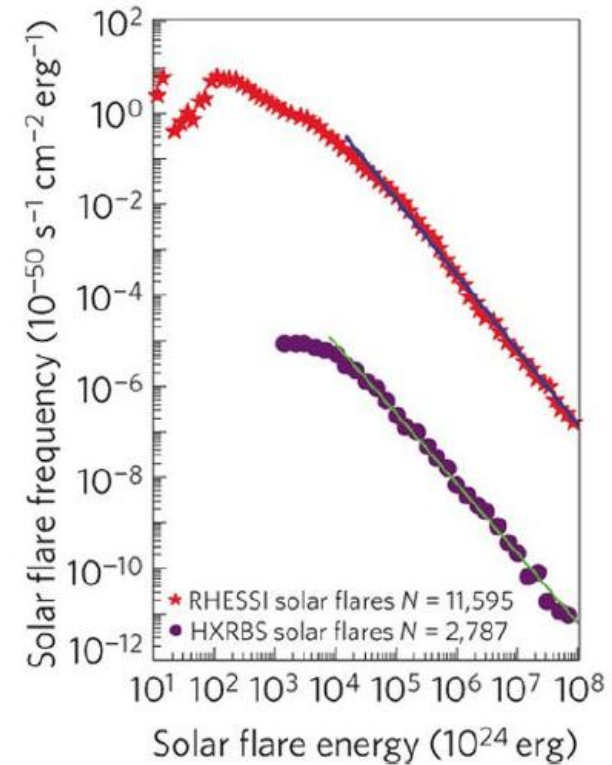
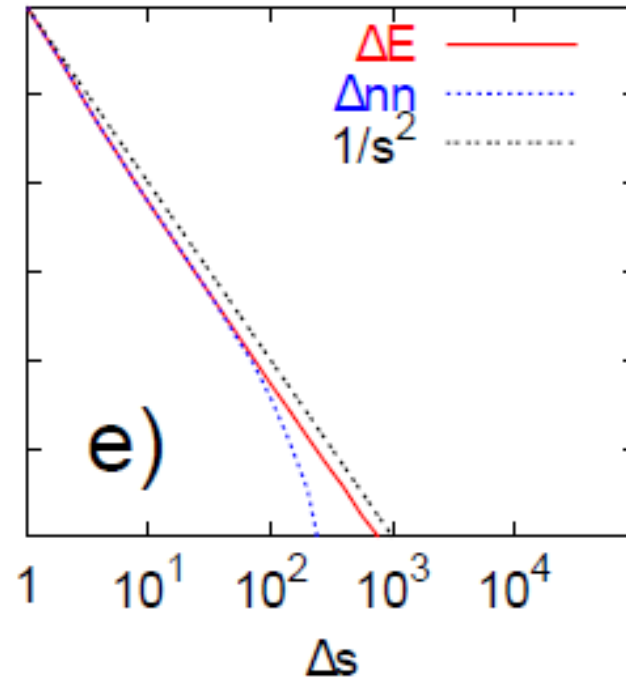
- 1) Merge i and j . There are now two possibilities:
 - a) If i and j have the same sign, all the links from i and j are assigned to the merged node. Thereby, the merged node has the same neighbors as i and j had together prior the merging, see Fig. 1.37(a).
 - b) If i and j are of opposite sign, the resulting vertex is assigned the sign of the sum $q_i + q_j$. Thereby, a number $\max\{|q_i|, |q_j|\} - |q_i + q_j|$ of links are annihilated in such a way that only the two merging nodes change their number of links. This is done by reconnecting donor nodes of incoming links to acceptor nodes of outgoing links, see Fig. 1.37(b).
- 2) One new vertex is created of random sign, with one edge being connected to a randomly chosen vertex.



Simplify as we add nodes and not reconnect loops, perhaps wrong

$$P(k) \propto 1/k^2. \quad P_{\Delta}(\Delta k) \propto 1/\Delta k^3.$$

Model $\leftrightarrow$ solar flare bursts:



$$P_{\Delta}(\Delta k) \propto 1/\Delta k^3$$

$$P(E) \propto 1/E^{1.65}$$

Thus model too steep, while data indicate average **dominated by largest event**

Note; First Return Of The Random Walk

- Cluster size distribution in Bethe lattice (ch. 3)
- Critical branching (Uranium Fission, ch. 4)
- In SOC: Sandpile and evolution models (ch. 4)
- Merging and Injection (ch. 6)
- And we will see it again in Econophysics (strike price distribution), ch. 7

→ So please understand origin of $1/t^{1.5}$